

BFS Traversal

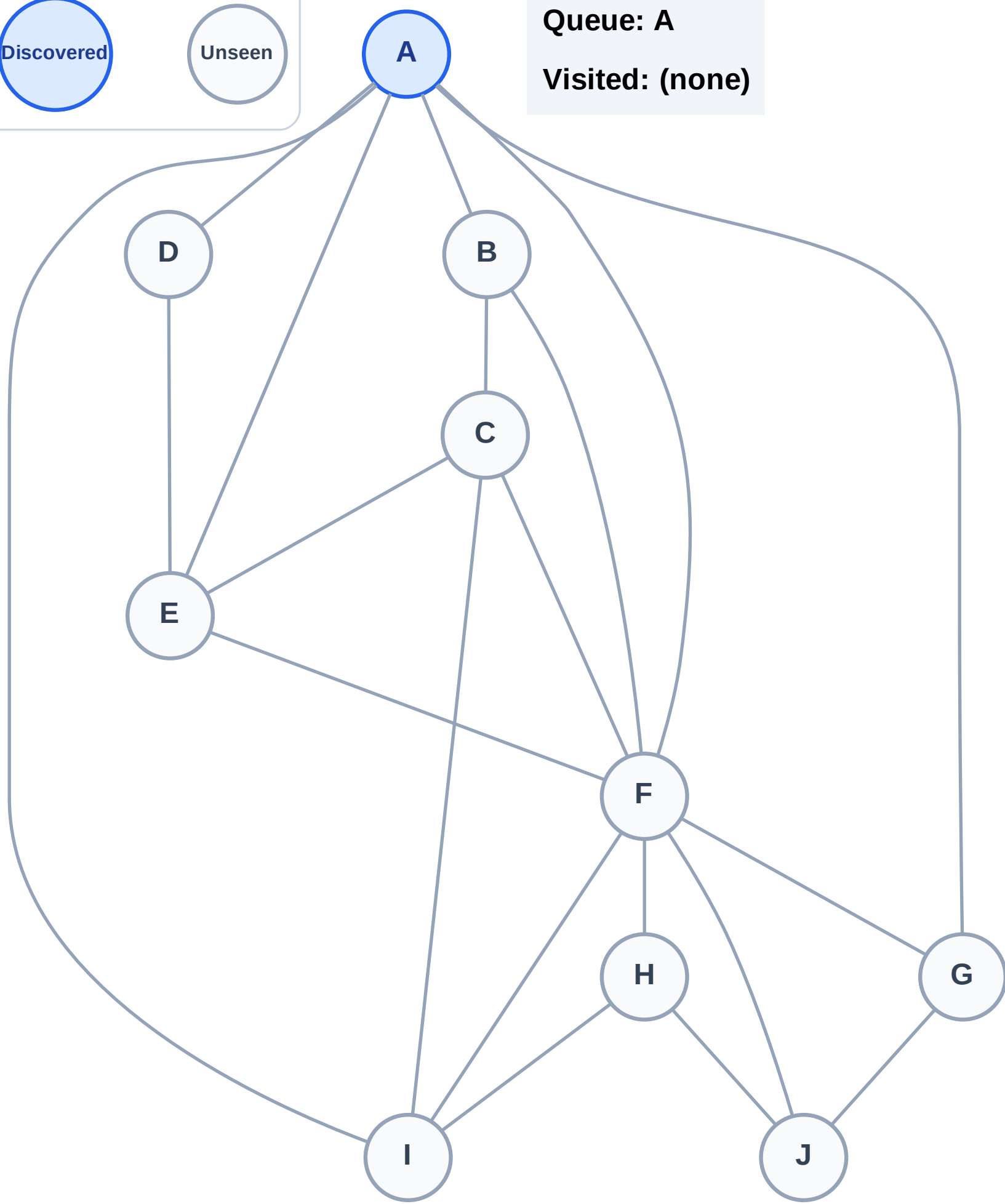
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

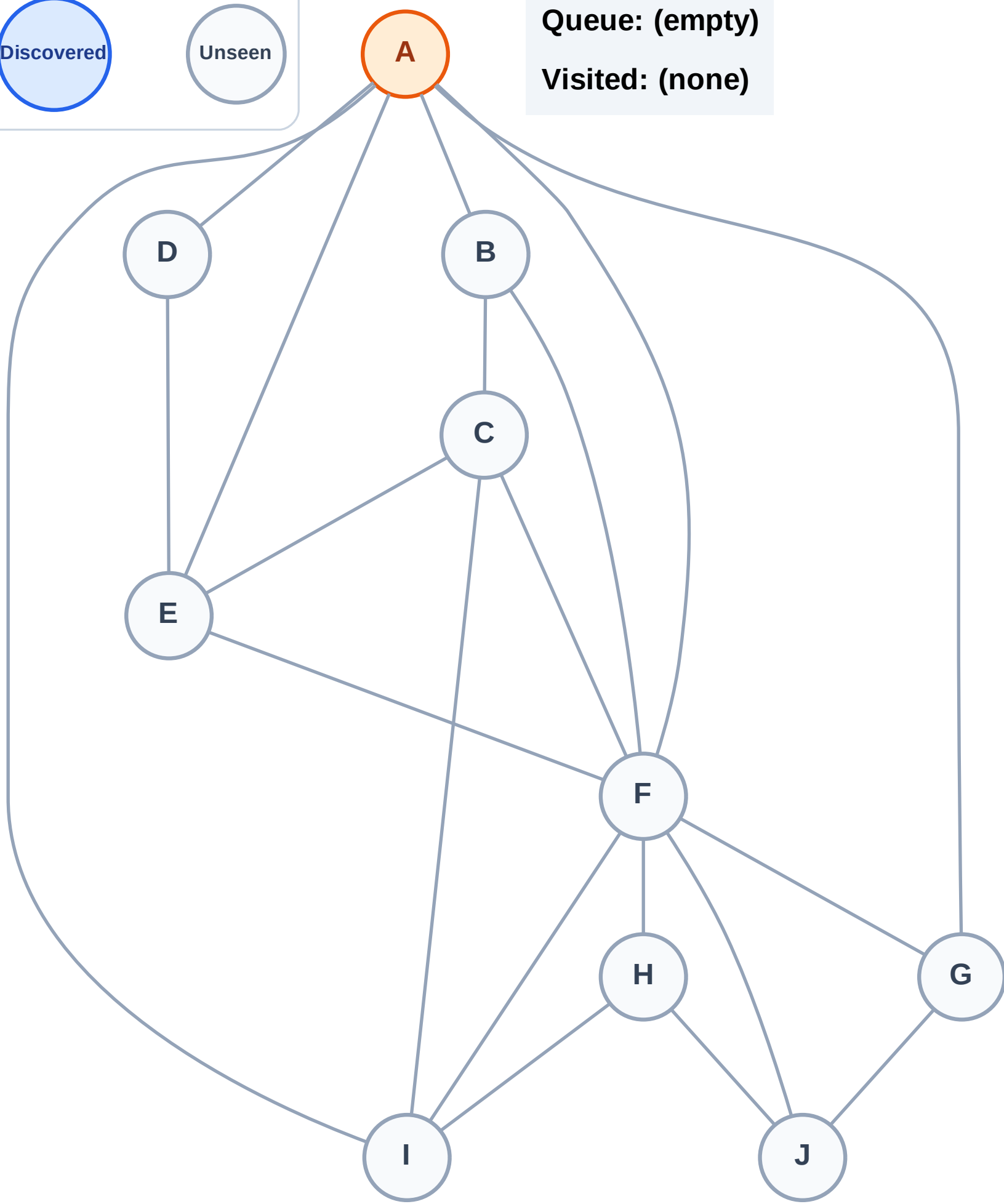
Step 2: Process node A

Legend



Queue: (empty)

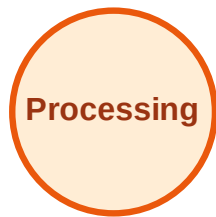
Visited: (none)



BFS Traversal

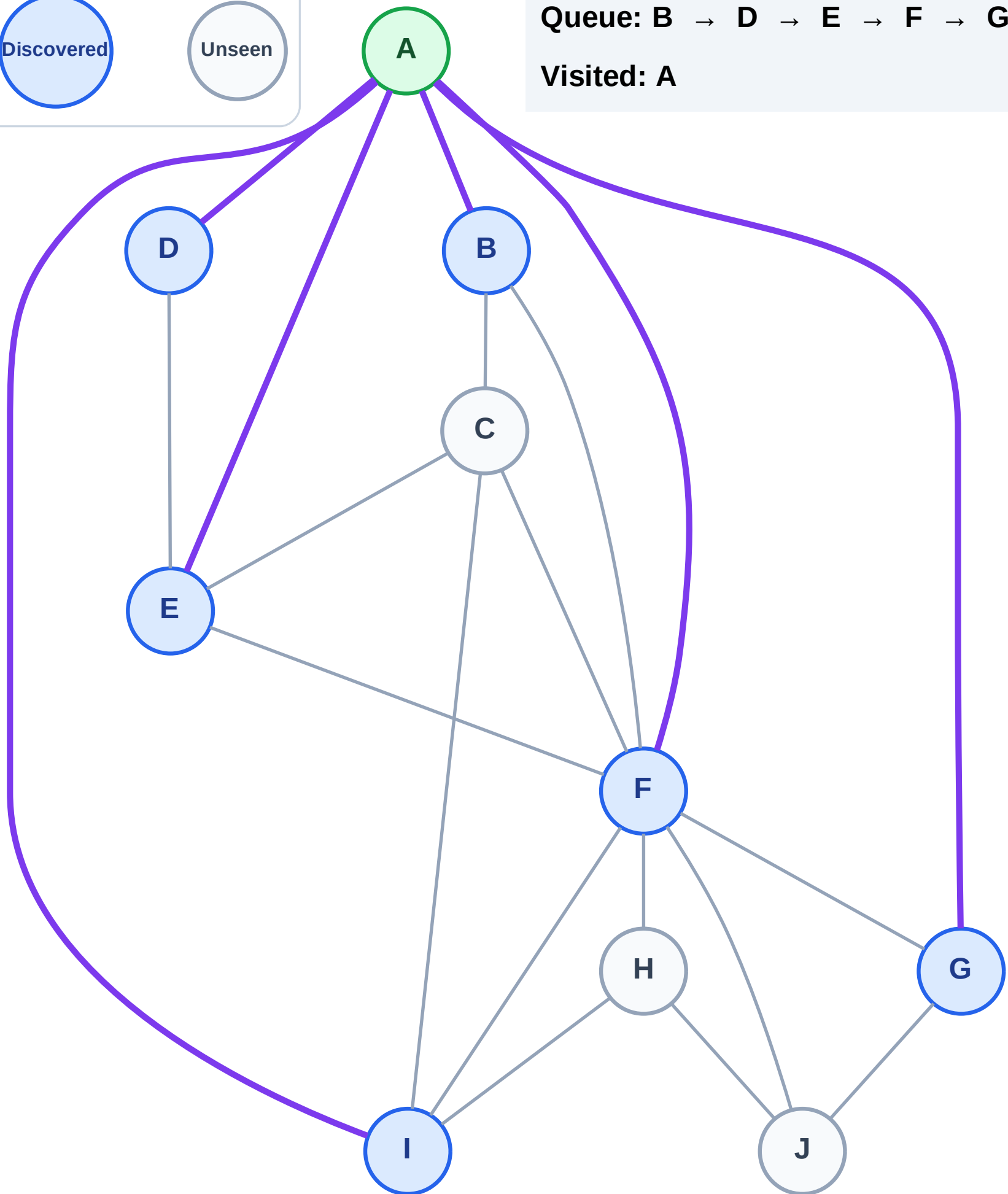
Step 3: Discover B, D, E, F, G, I from A

Legend



Queue: B → D → E → F → G → I

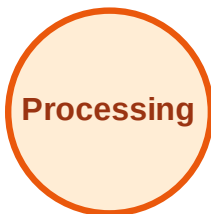
Visited: A



BFS Traversal

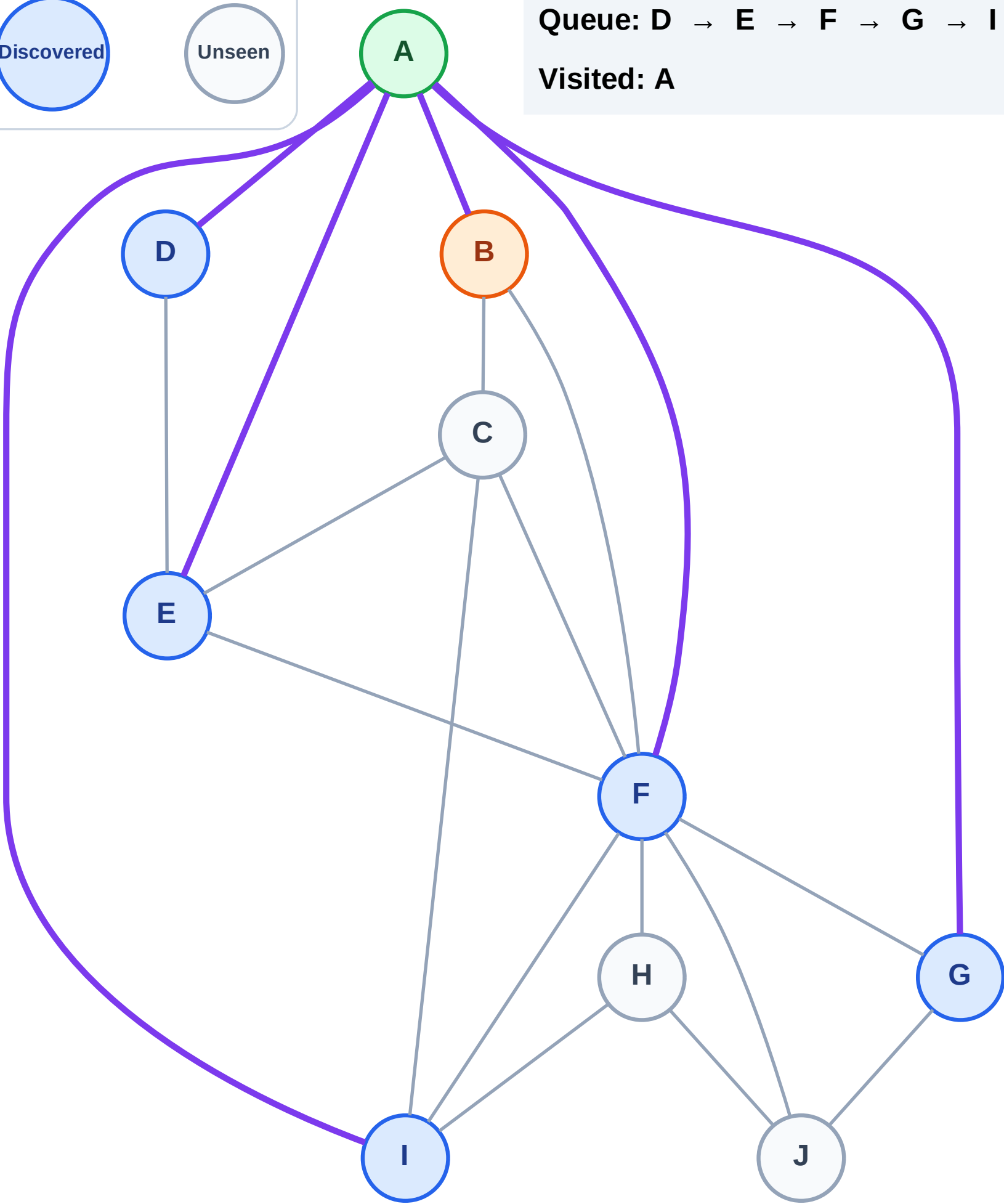
Step 4: Process node B

Legend



Queue: D → E → F → G → I

Visited: A



BFS Traversal

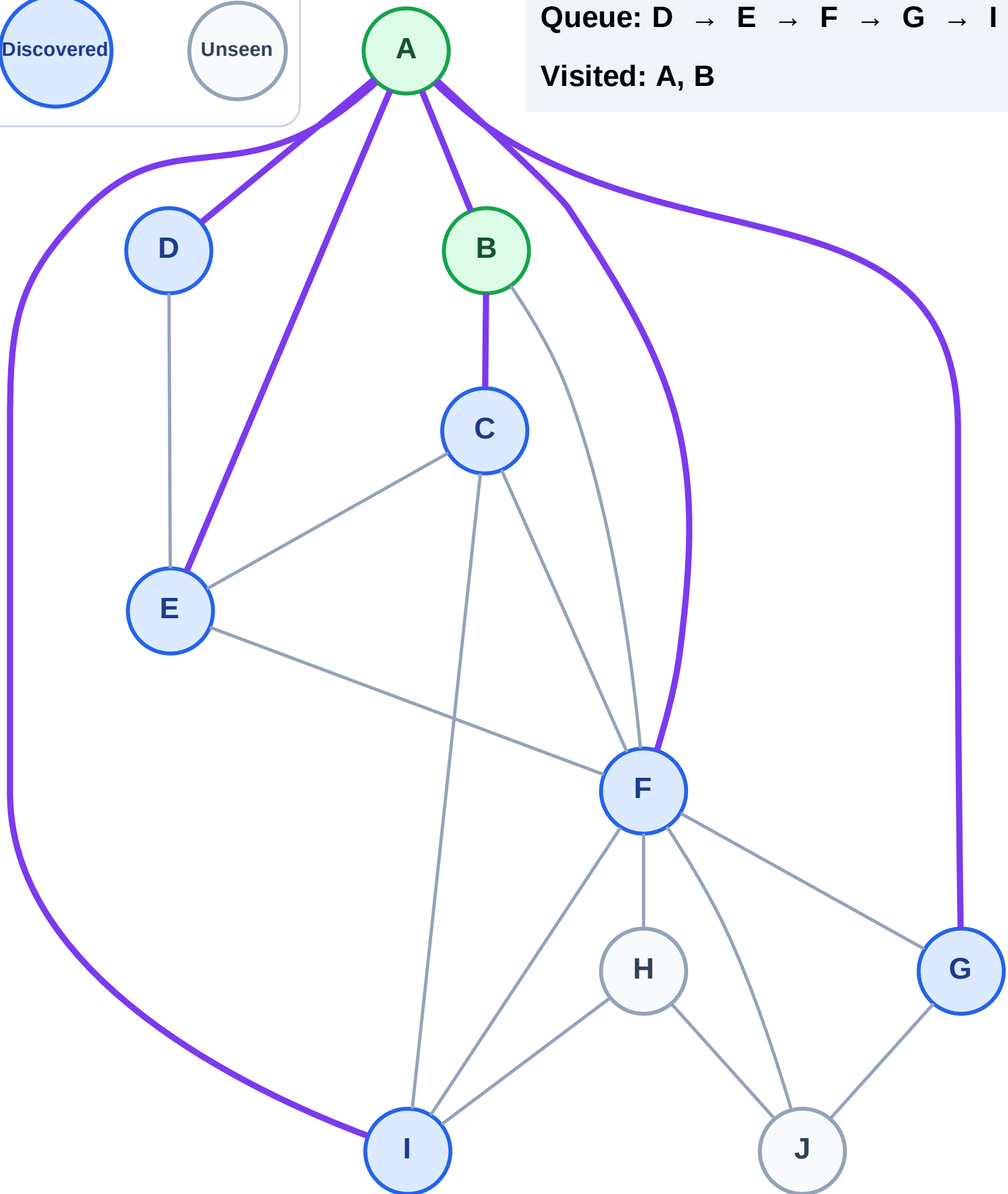
Step 5: Discover C from B

Legend



Queue: D → E → F → G → I → C

Visited: A, B



BFS Traversal

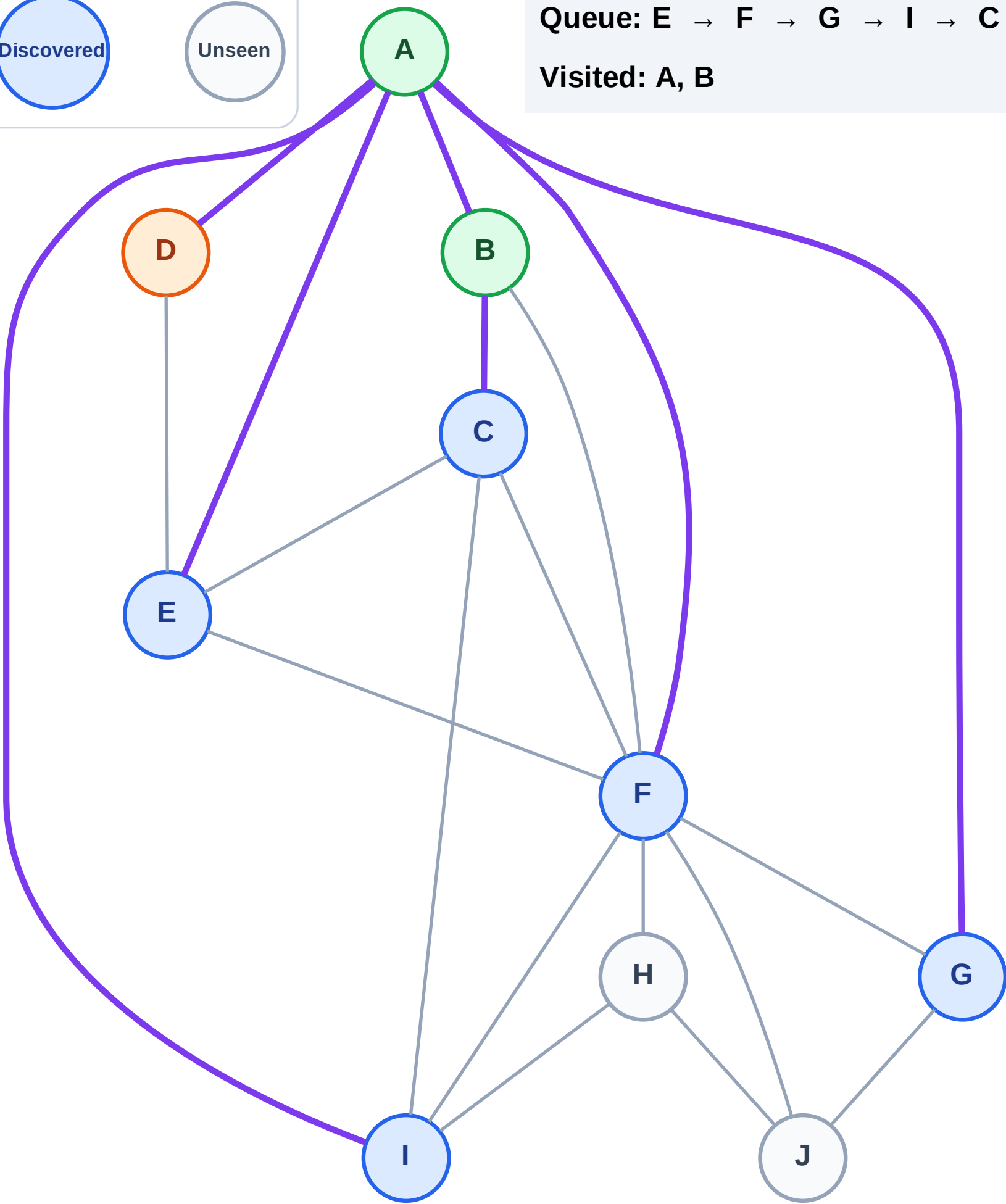
Step 6: Process node D

Legend



Queue: E → F → G → I → C

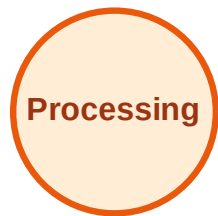
Visited: A, B



BFS Traversal

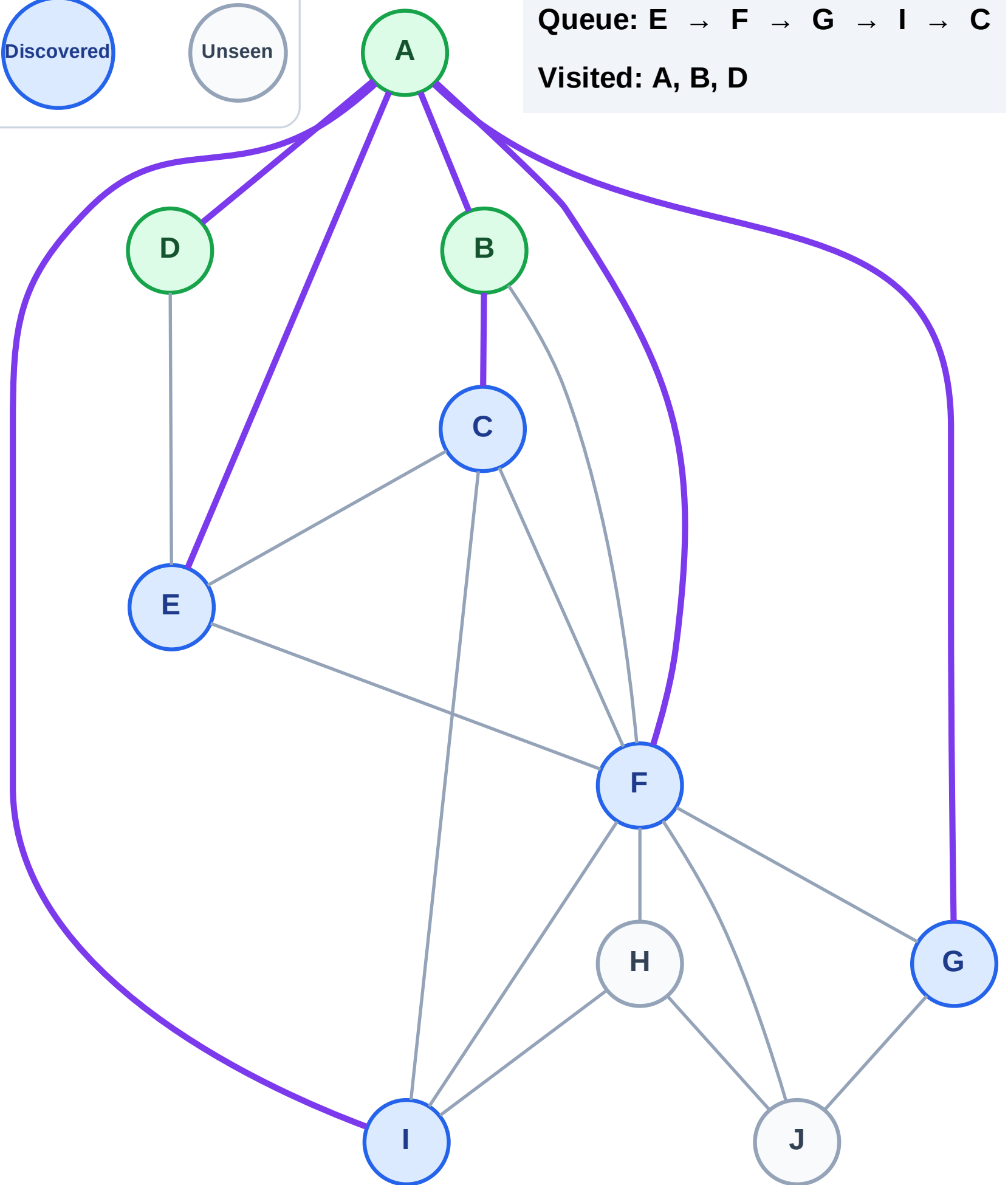
Step 7: No new neighbors from D

Legend



Queue: E → F → G → I → C

Visited: A, B, D



BFS Traversal

Step 8: Process node E

Legend

Complete

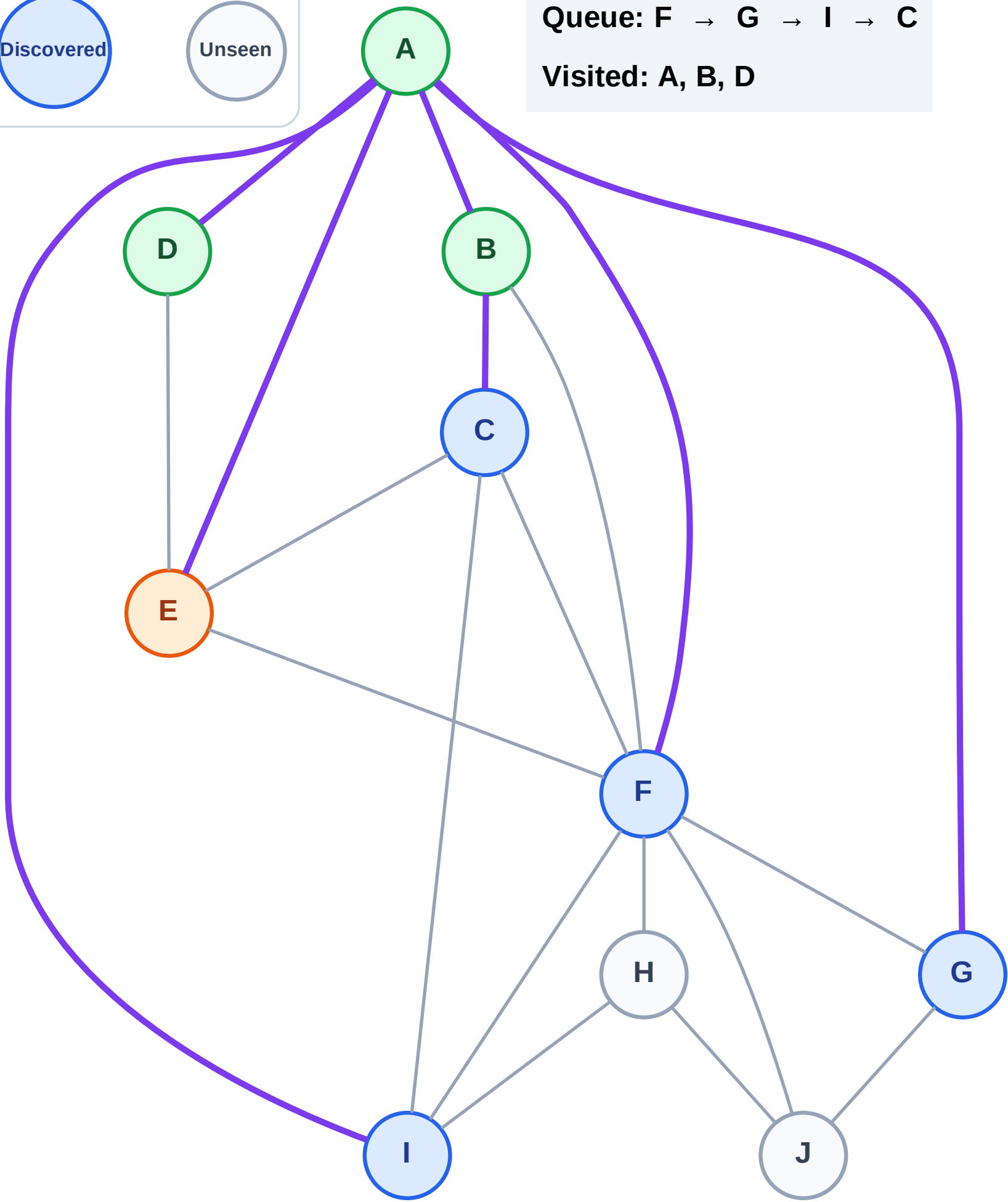
Processing

Discovered

Unseen

Queue: F → G → I → C

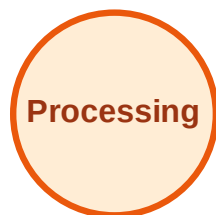
Visited: A, B, D



BFS Traversal

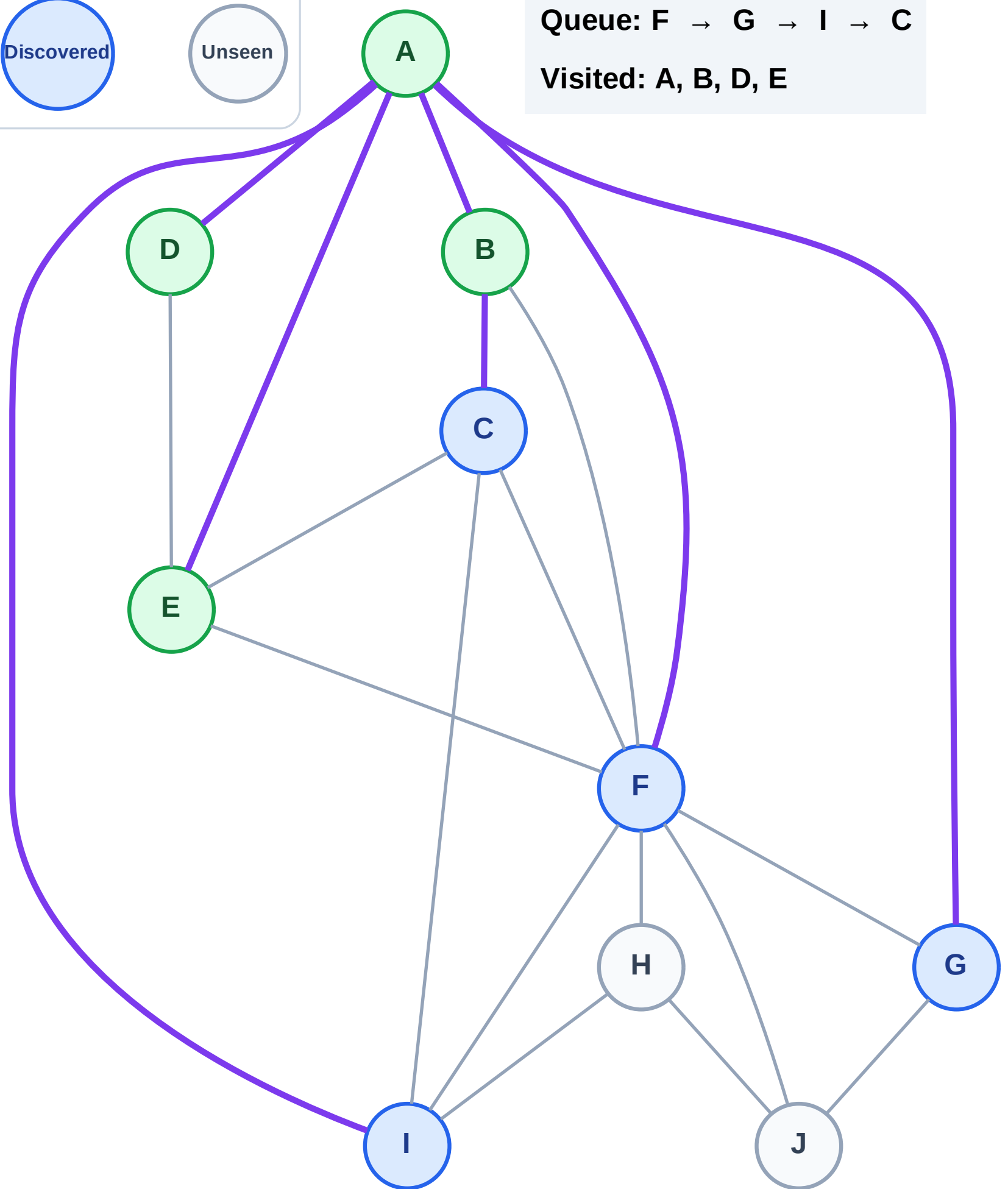
Step 9: No new neighbors from E

Legend



Queue: F → G → I → C

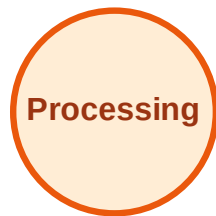
Visited: A, B, D, E



BFS Traversal

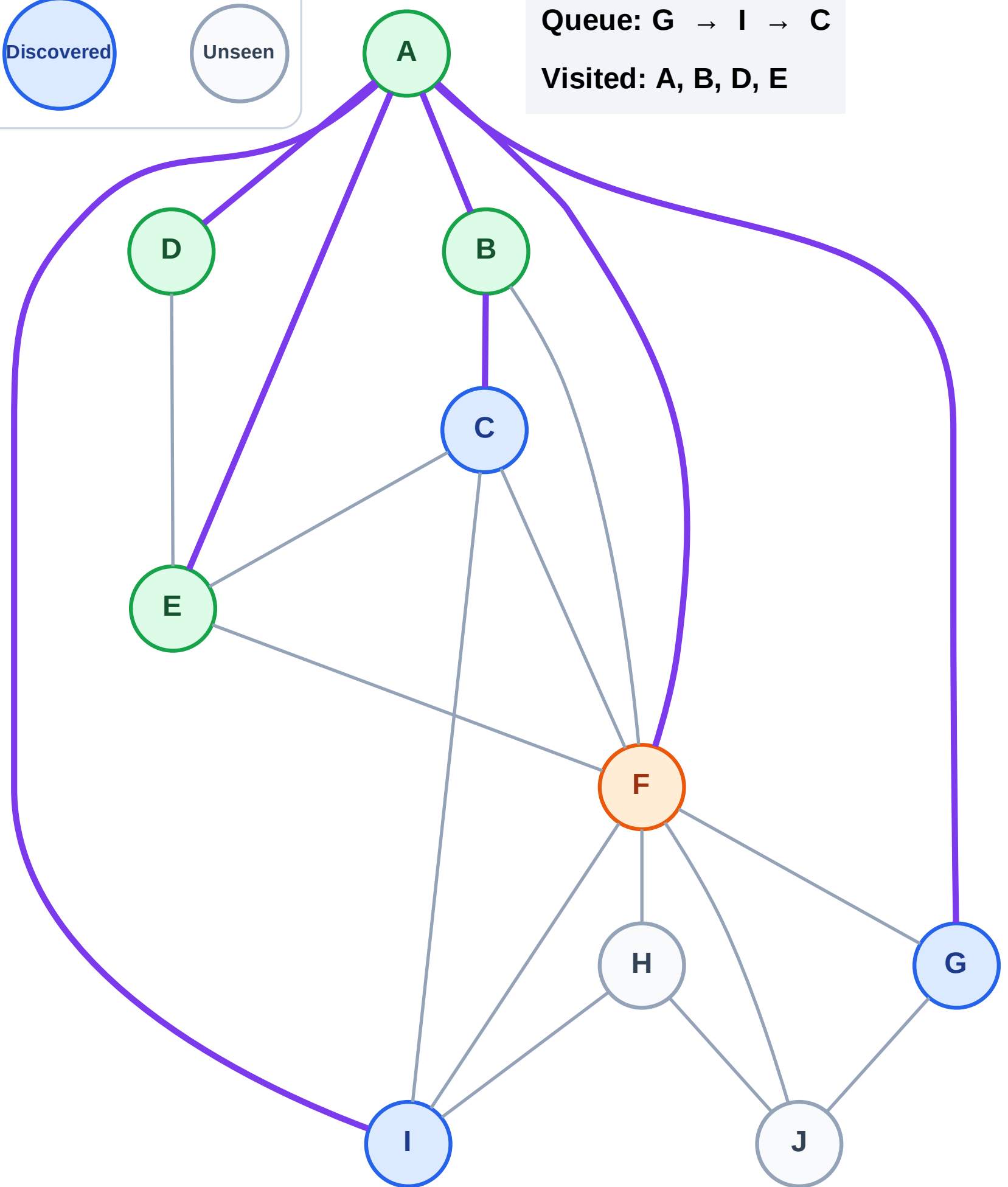
Step 10: Process node F

Legend



Queue: G → I → C

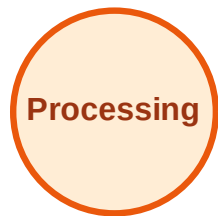
Visited: A, B, D, E



BFS Traversal

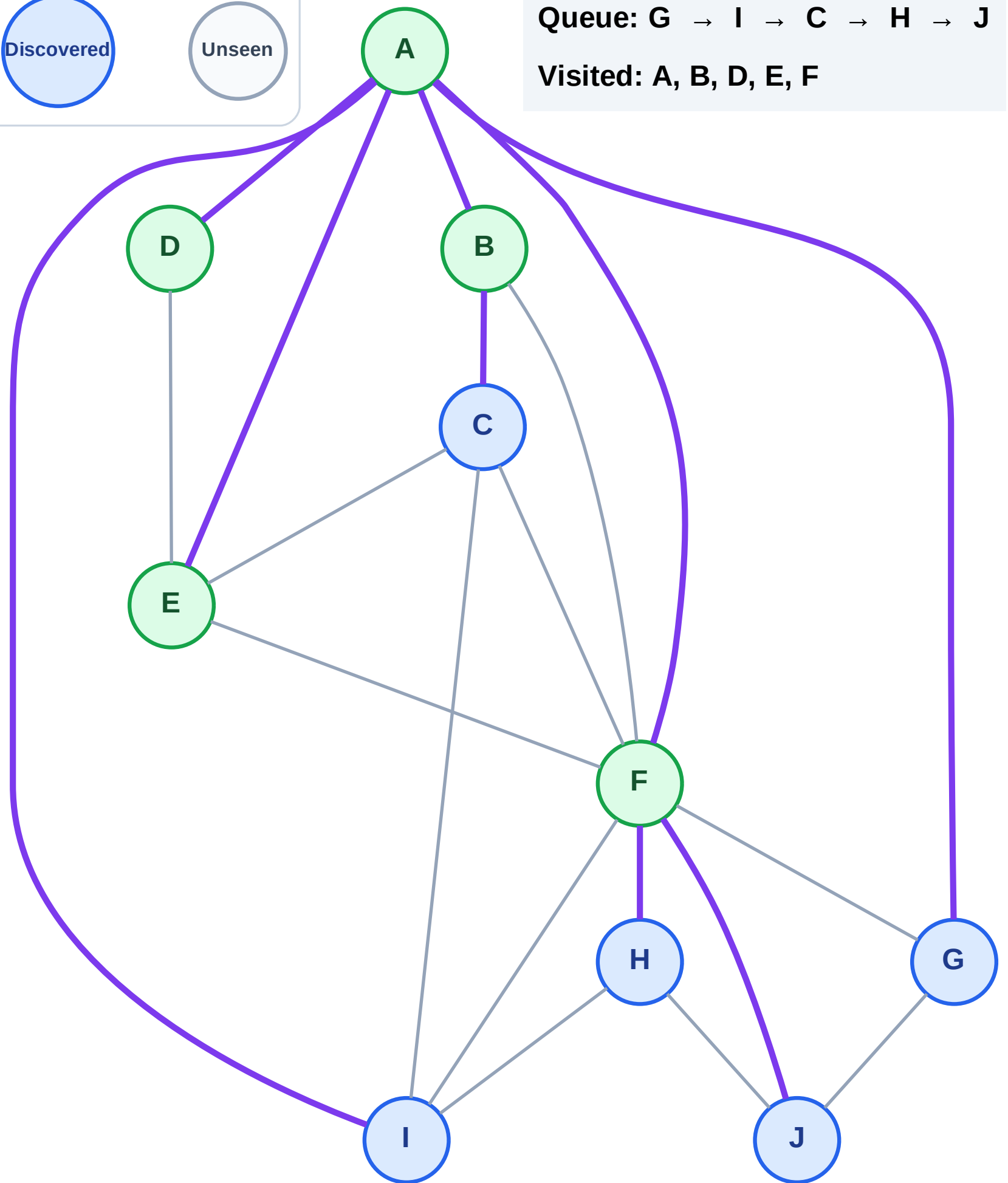
Step 11: Discover H, J from F

Legend



Queue: G → I → C → H → J

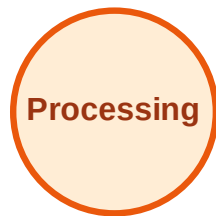
Visited: A, B, D, E, F



BFS Traversal

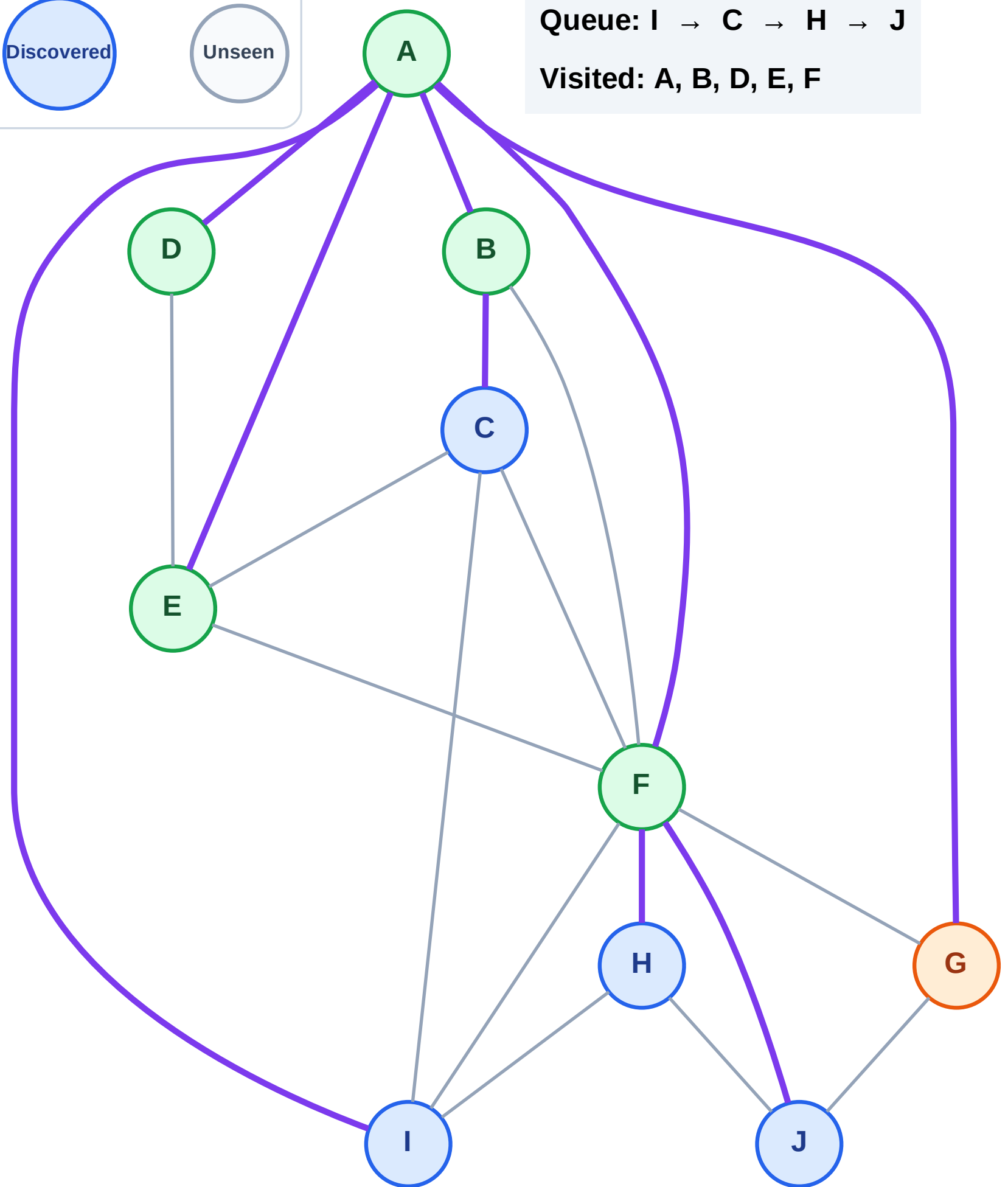
Step 12: Process node G

Legend



Queue: I → C → H → J

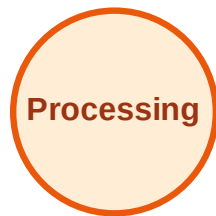
Visited: A, B, D, E, F



BFS Traversal

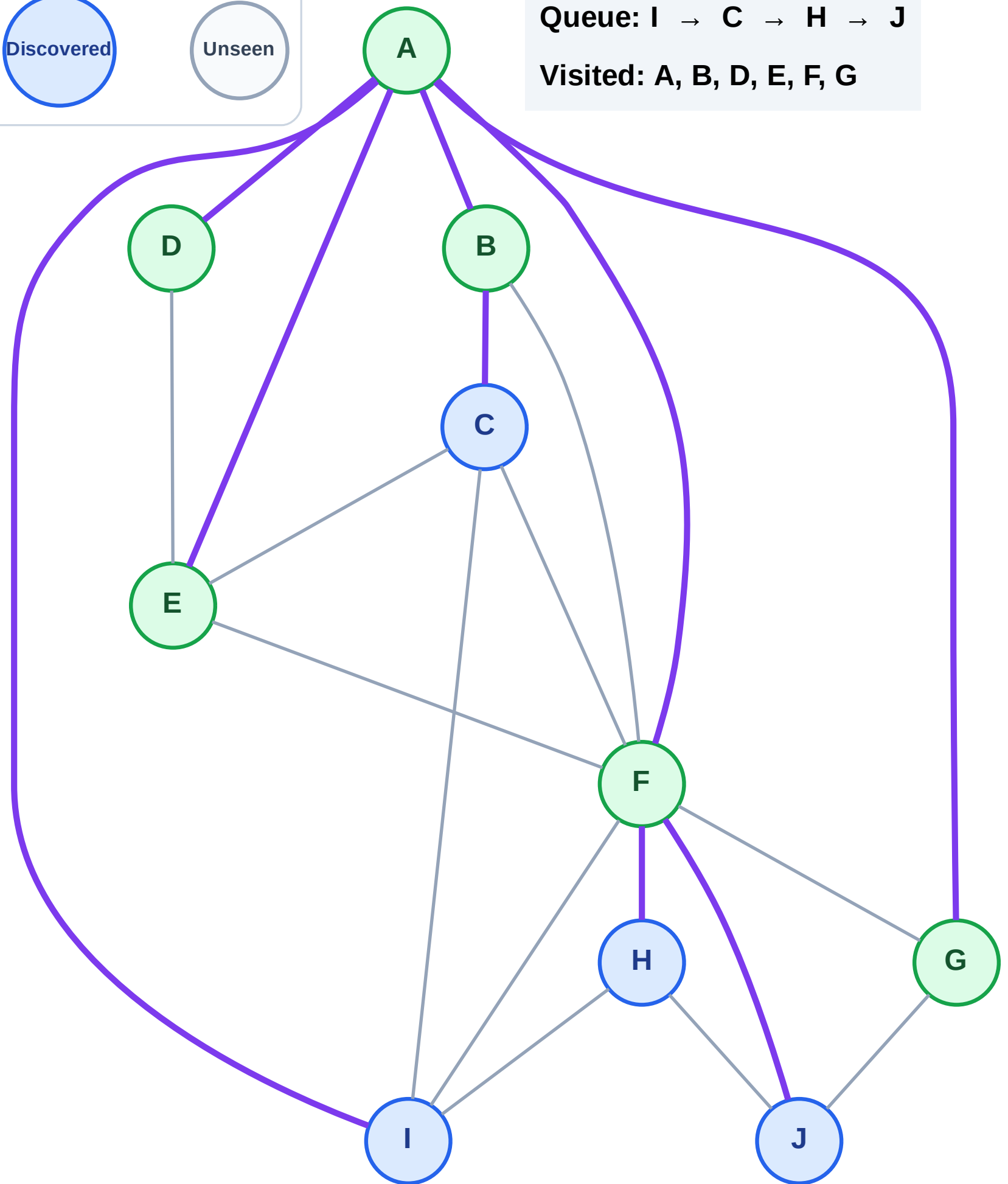
Step 13: No new neighbors from G

Legend



Queue: I → C → H → J

Visited: A, B, D, E, F, G



BFS Traversal

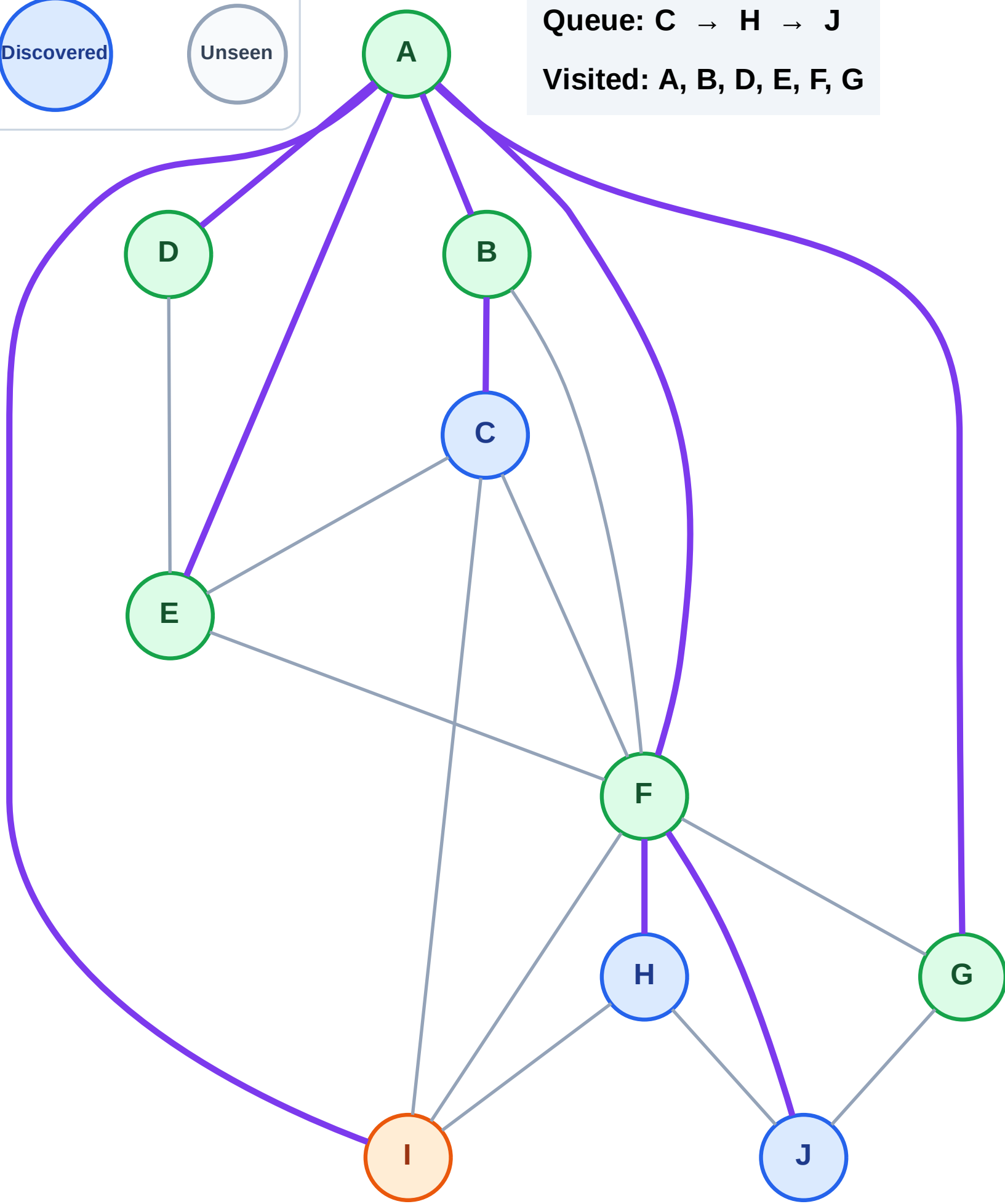
Step 14: Process node I

Legend



Queue: C → H → J

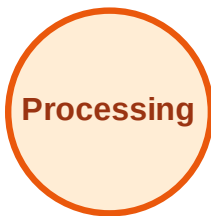
Visited: A, B, D, E, F, G



BFS Traversal

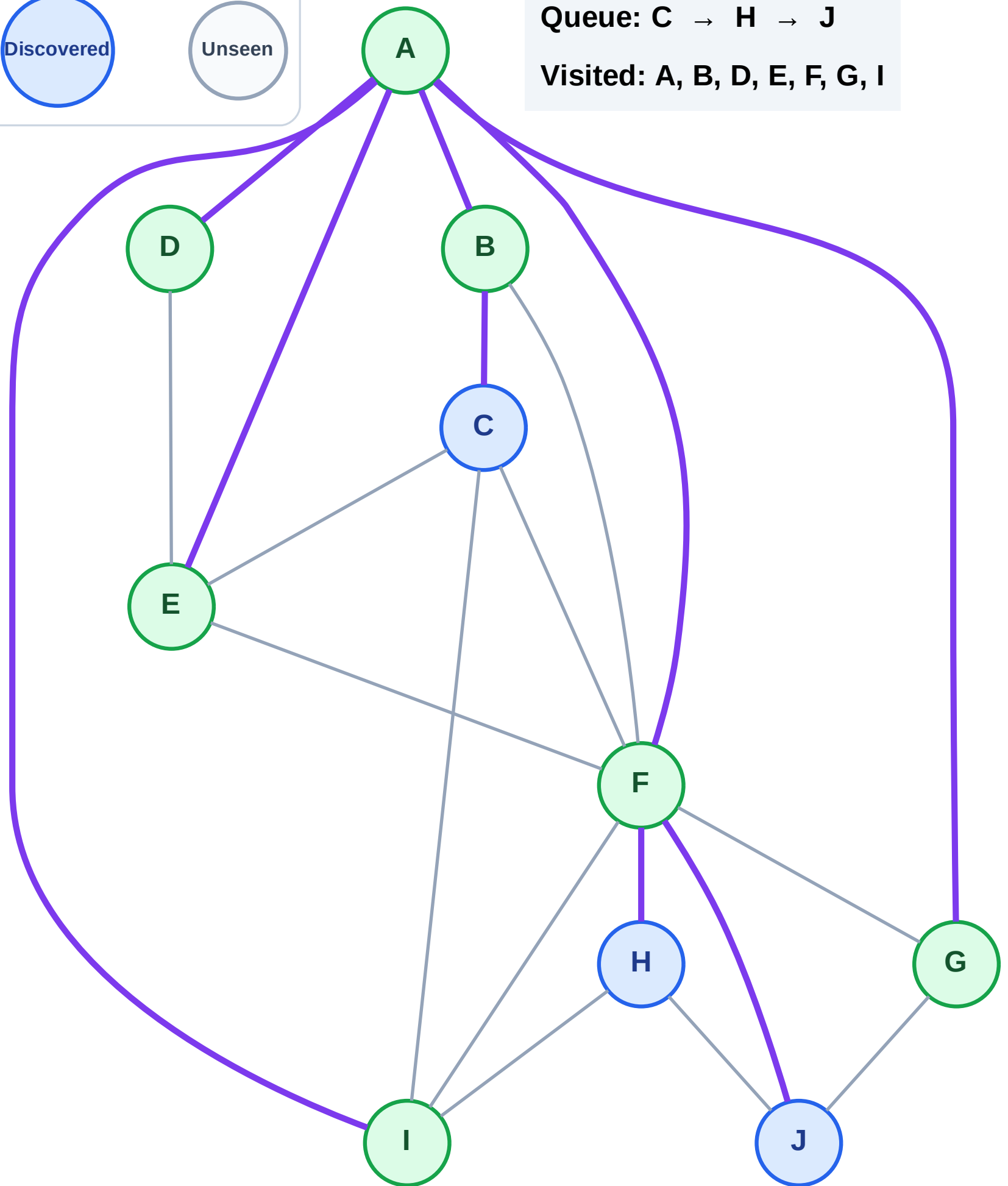
Step 15: No new neighbors from I

Legend



Queue: C → H → J

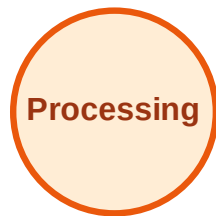
Visited: A, B, D, E, F, G, I



BFS Traversal

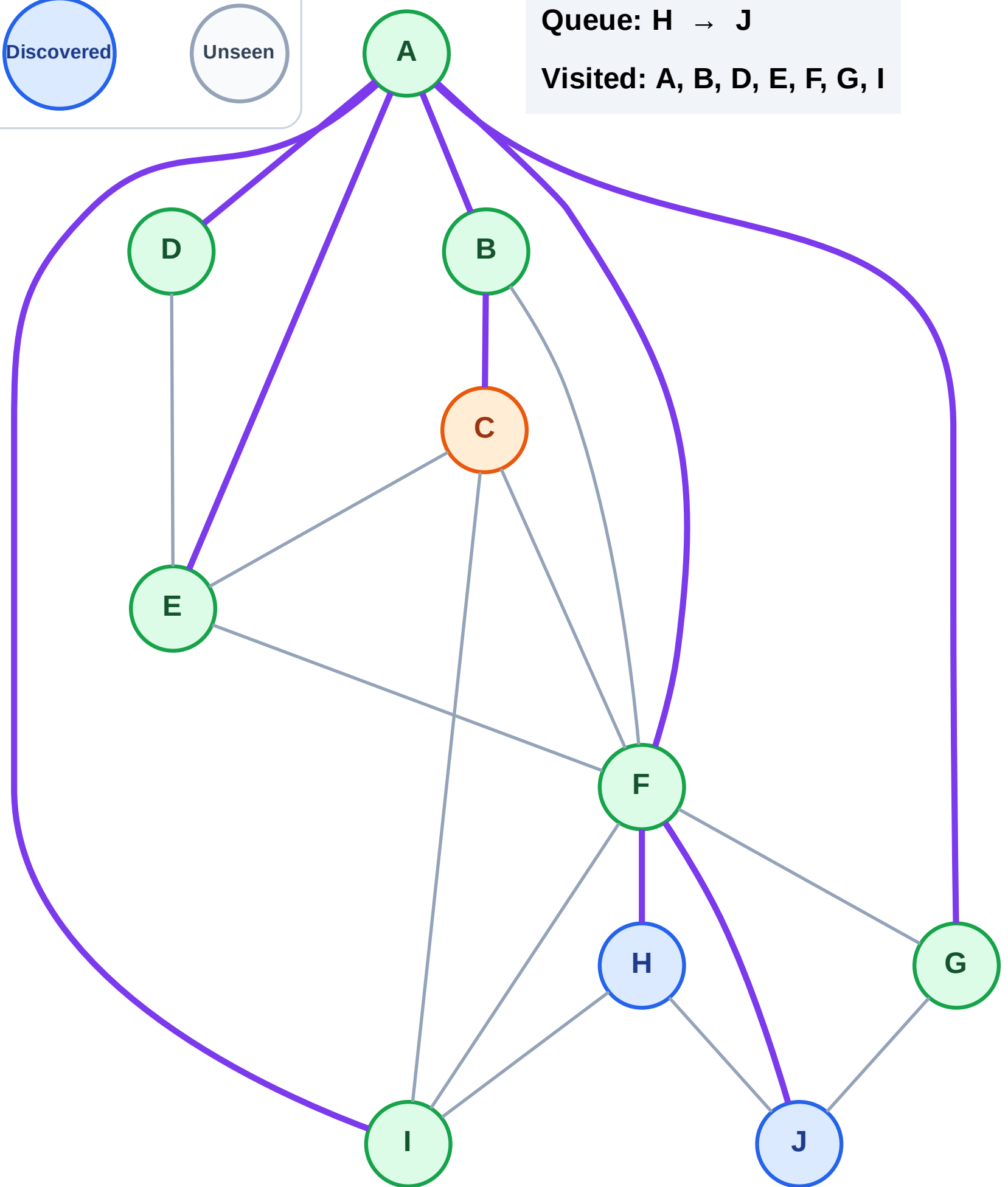
Step 16: Process node C

Legend



Queue: H → J

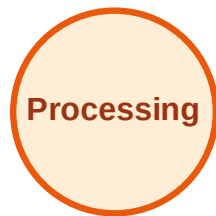
Visited: A, B, D, E, F, G, I



BFS Traversal

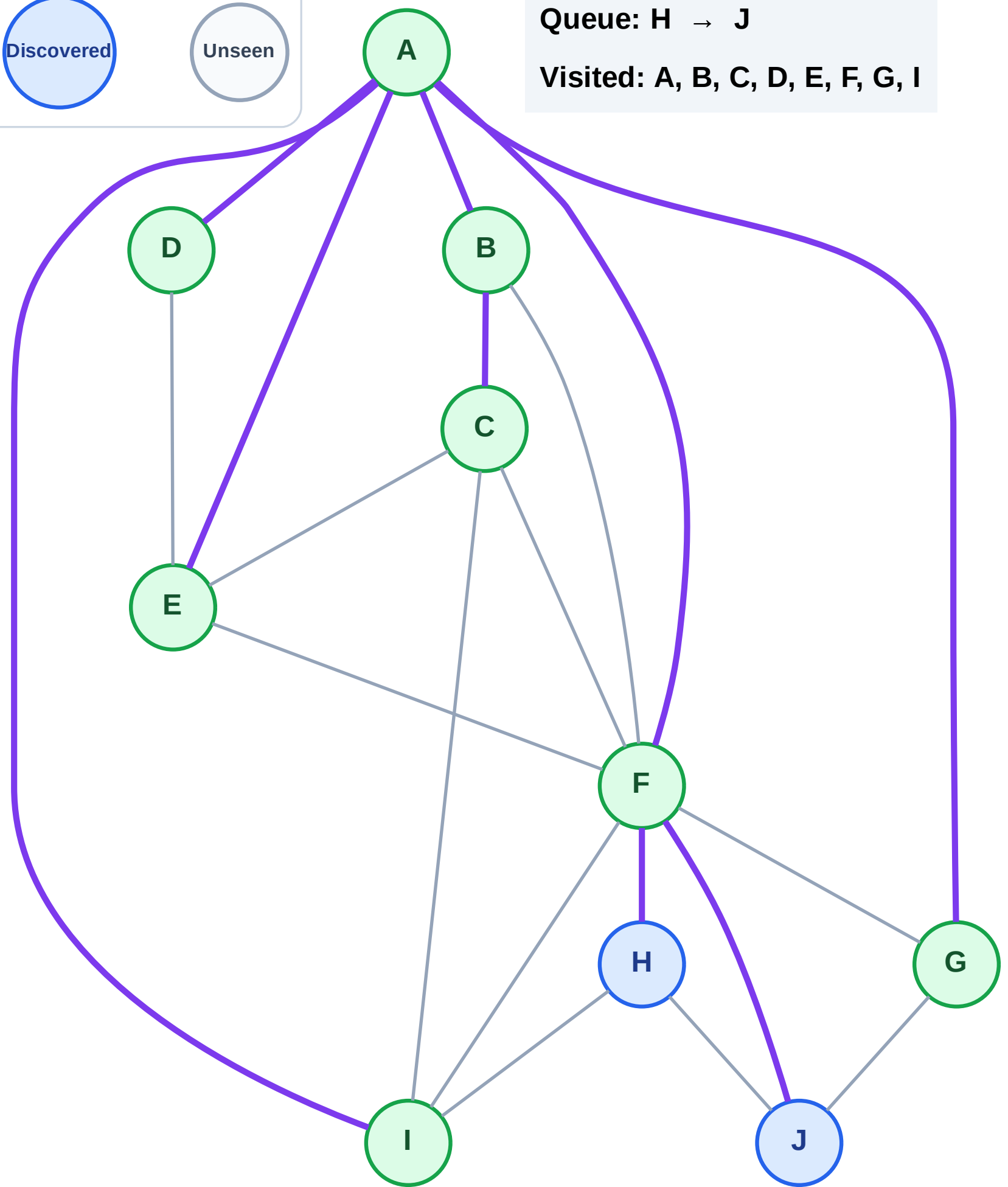
Step 17: No new neighbors from C

Legend



Queue: H → J

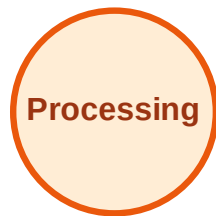
Visited: A, B, C, D, E, F, G, I



BFS Traversal

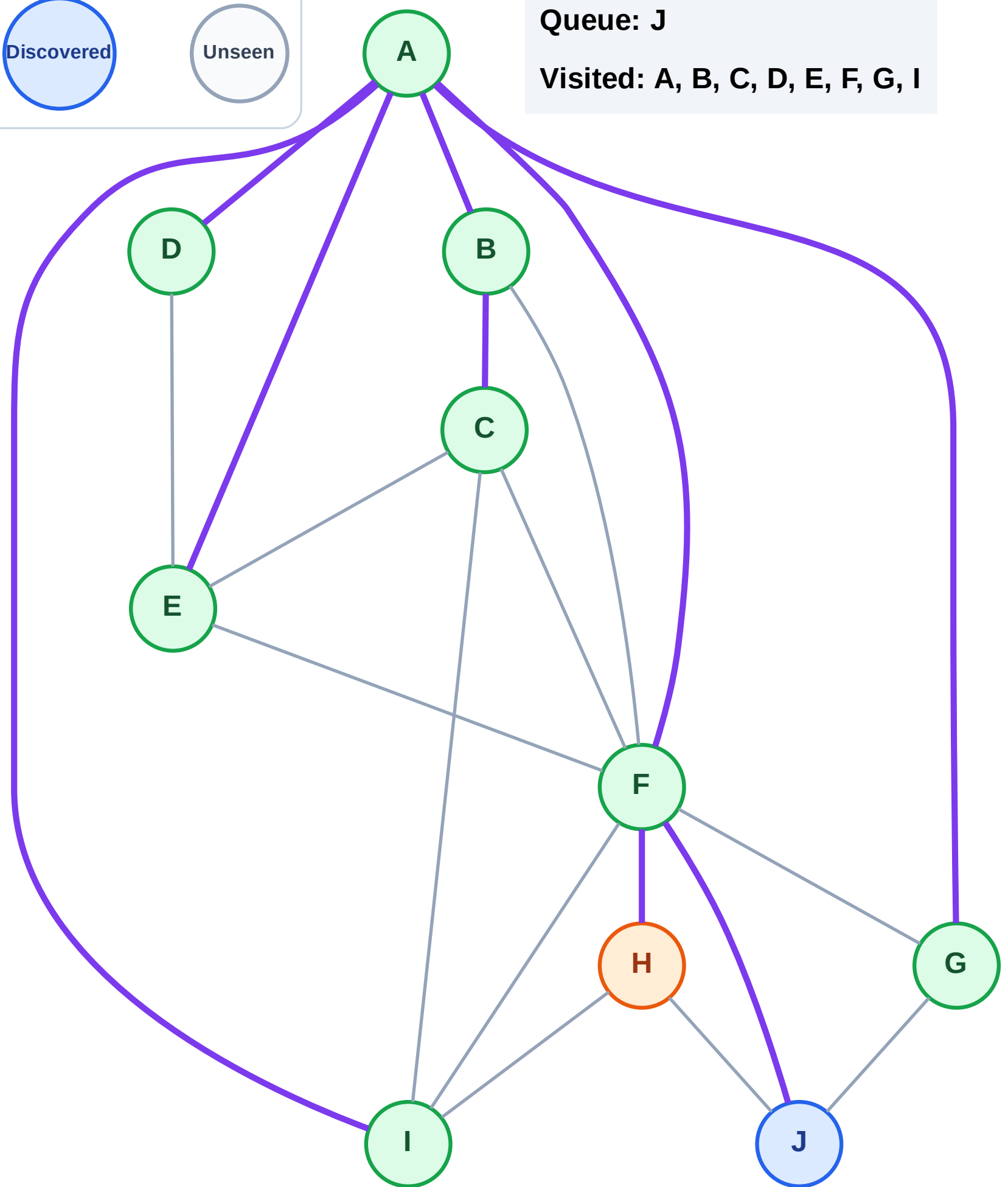
Step 18: Process node H

Legend



Queue: J

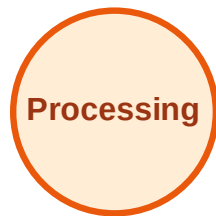
Visited: A, B, C, D, E, F, G, I



BFS Traversal

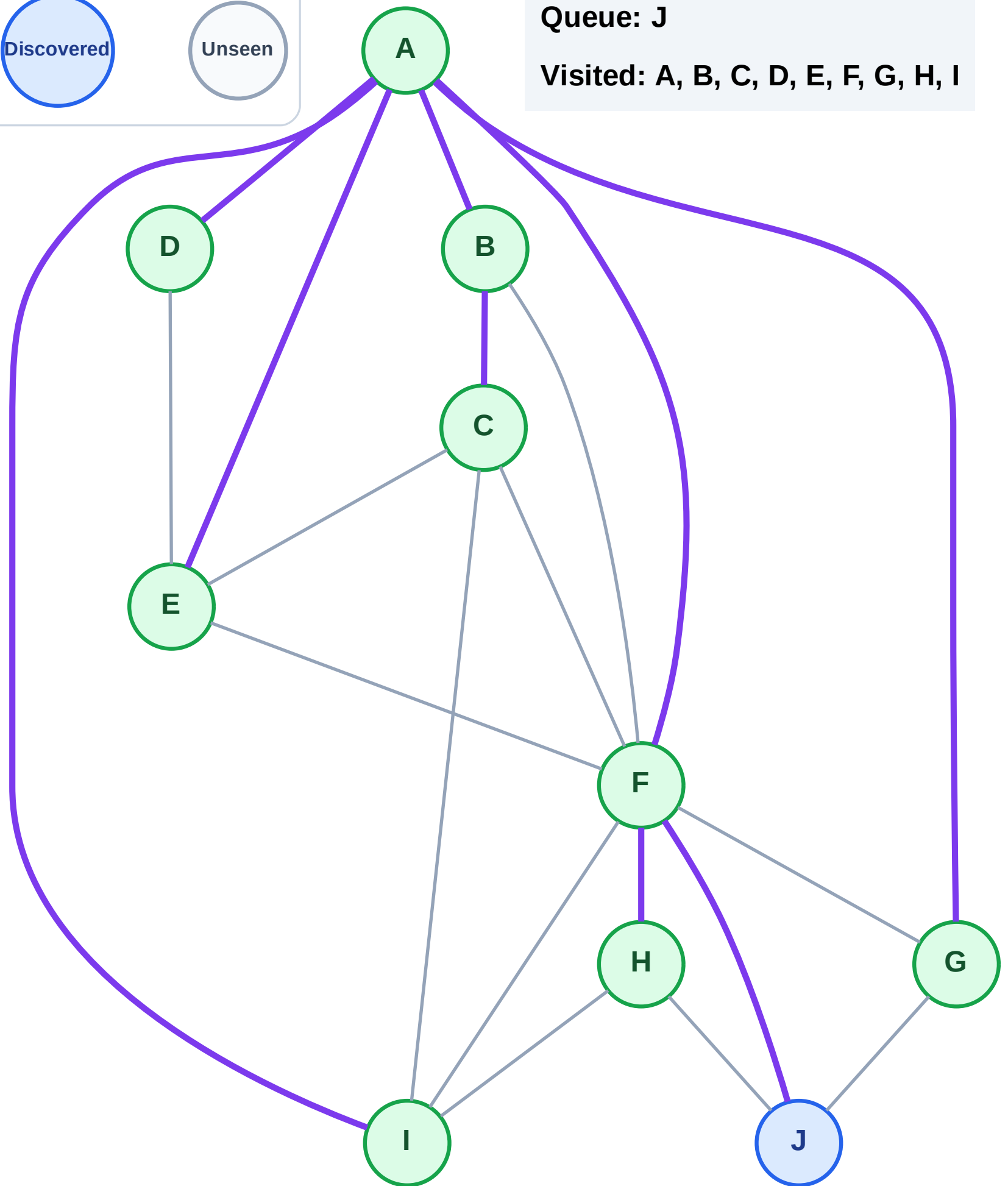
Step 19: No new neighbors from H

Legend



Queue: J

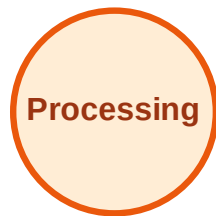
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

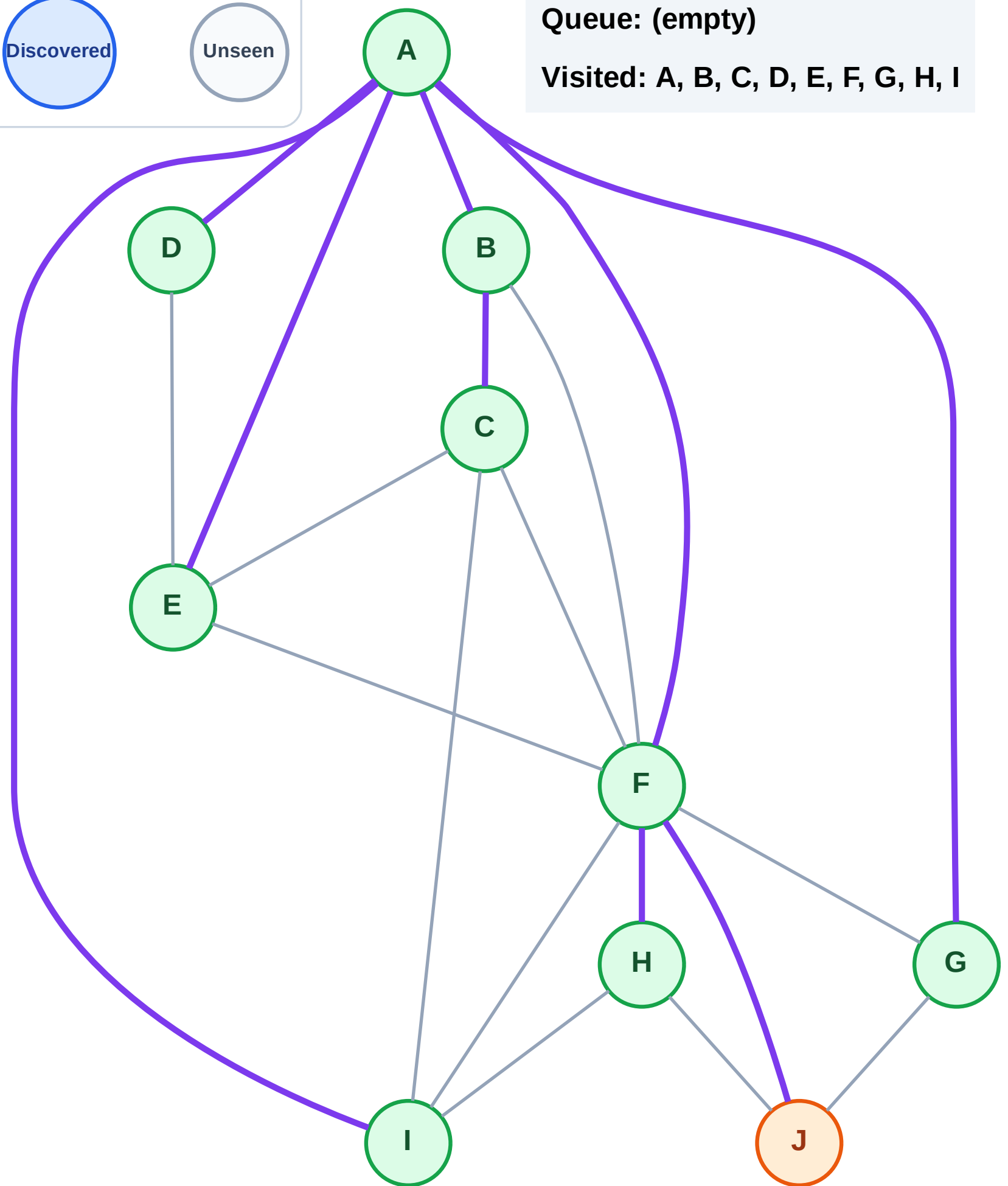
Step 20: Process node J

Legend



Queue: (empty)

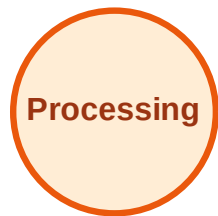
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

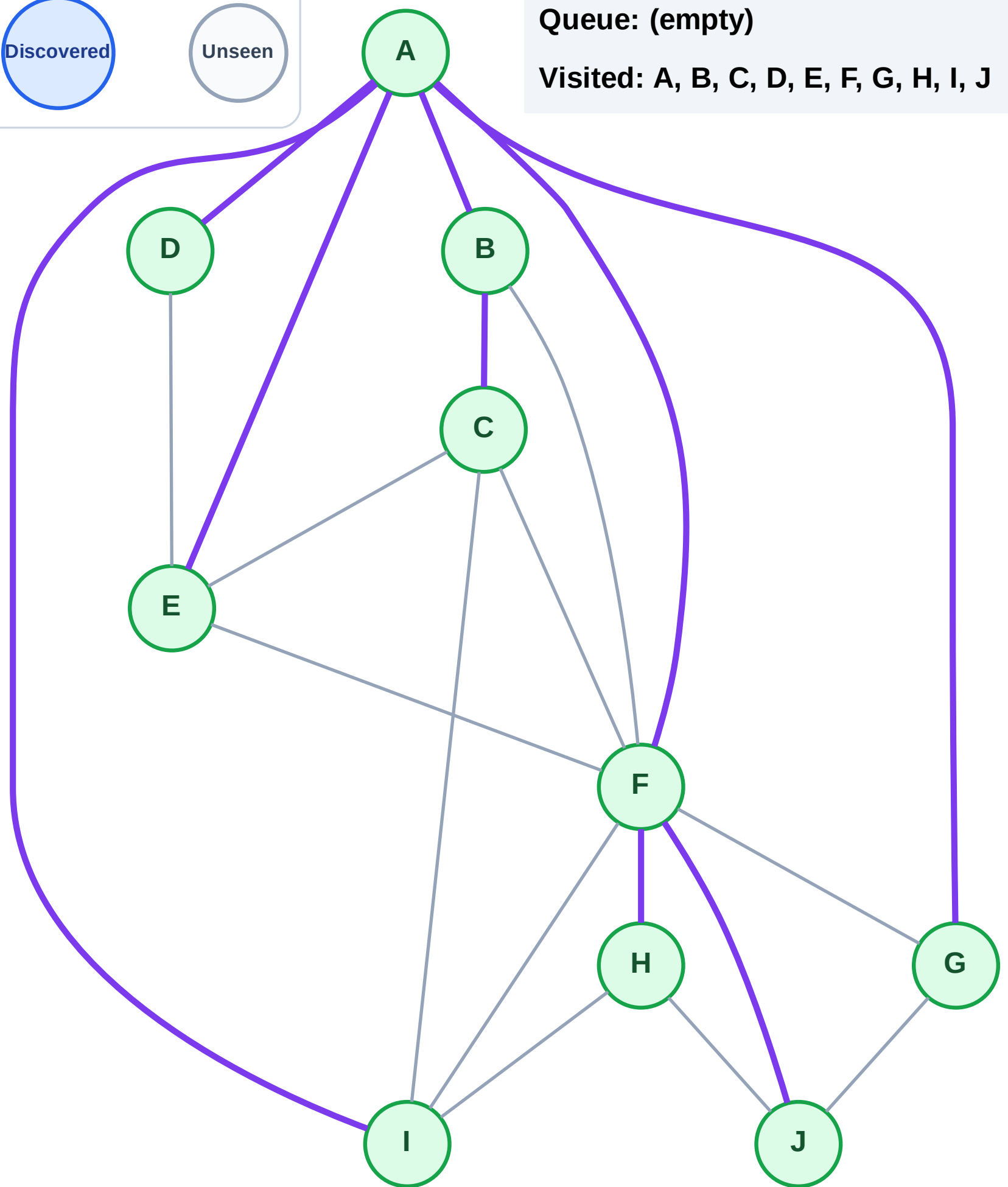
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

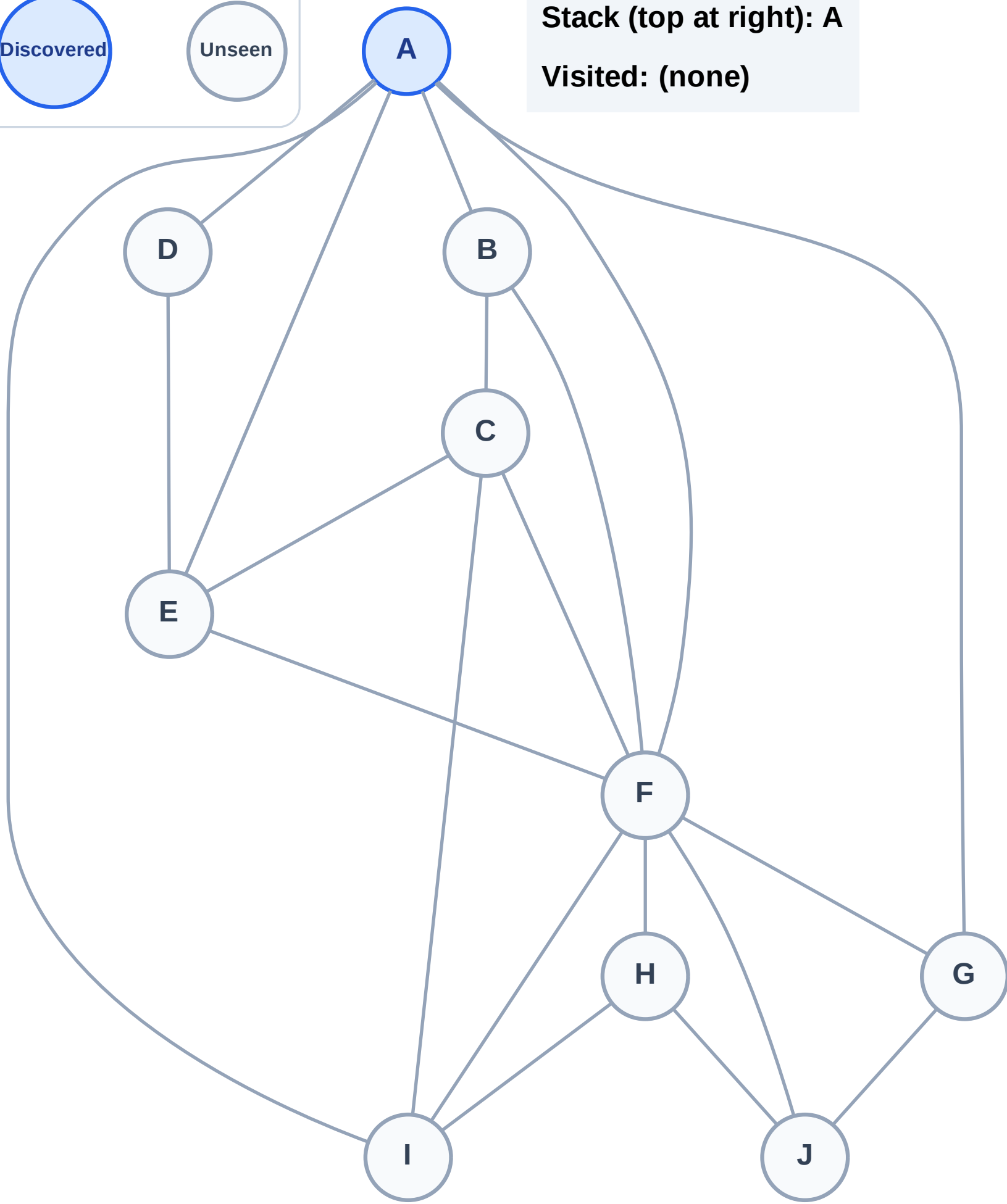
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

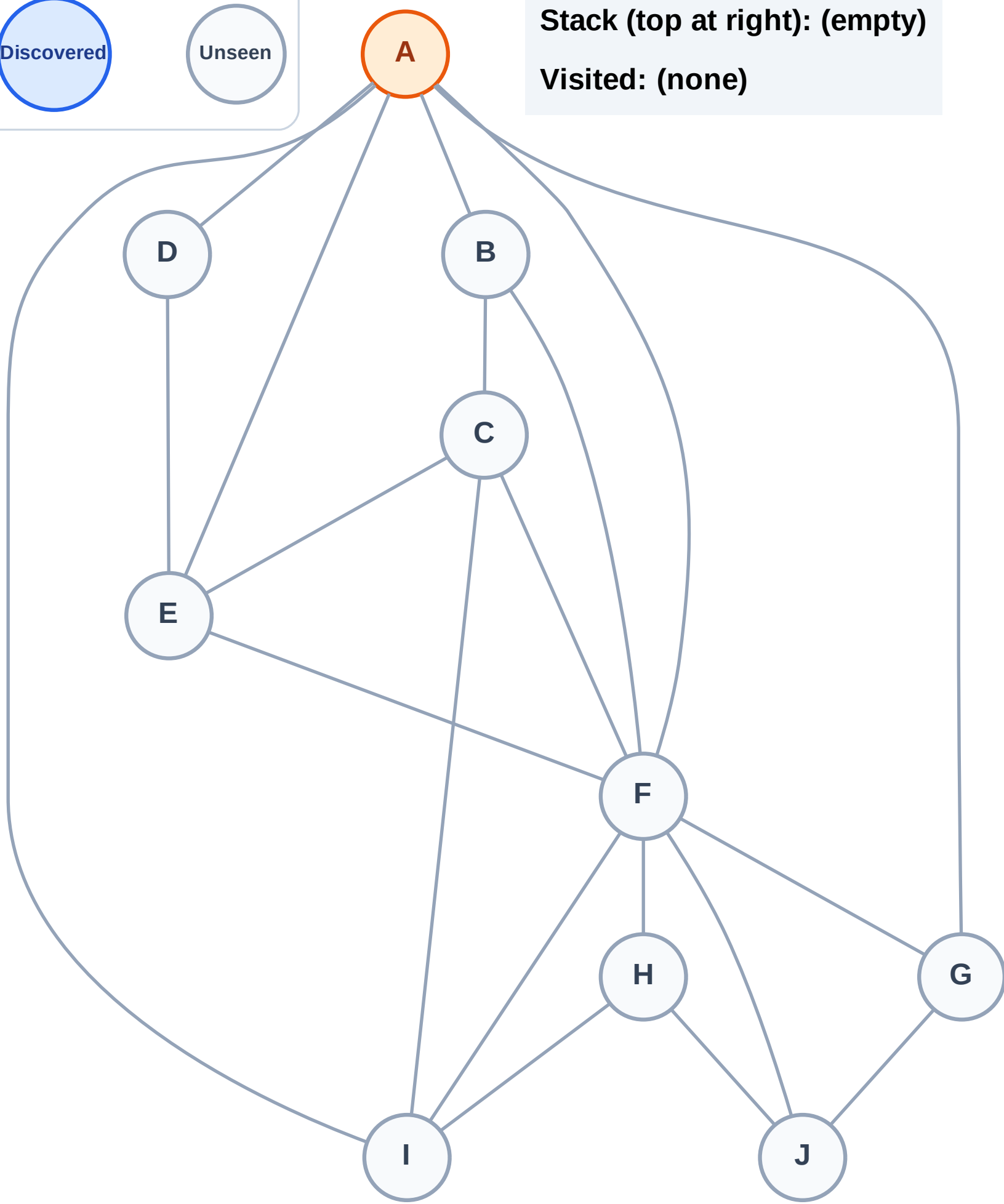
Step 2: Process node A

Legend



Stack (top at right): (empty)

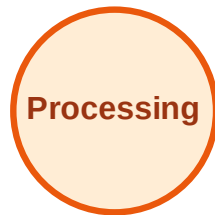
Visited: (none)



DFS Traversal

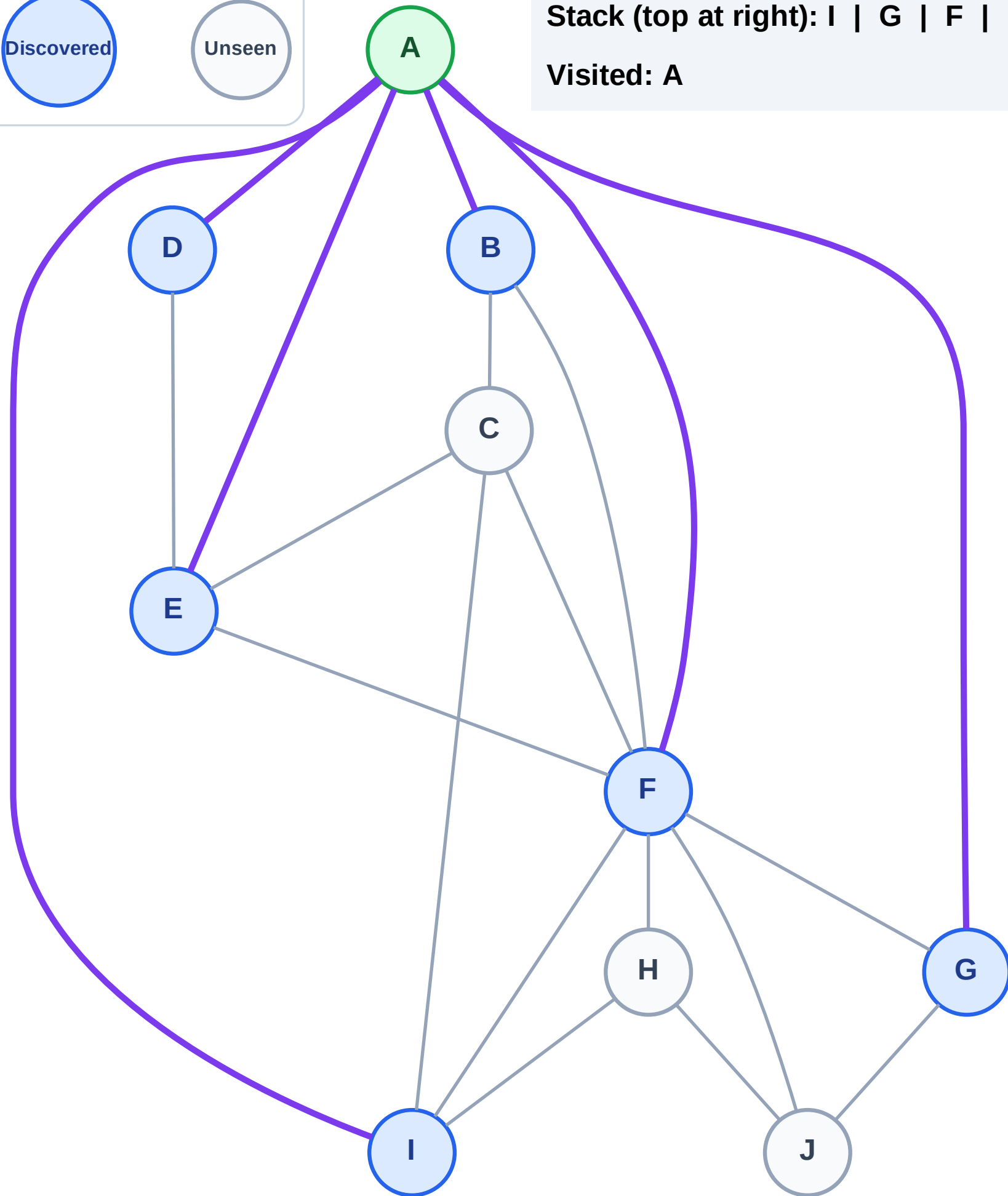
Step 3: Discover I, G, F, E, D, B from A

Legend



Stack (top at right): I | G | F | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

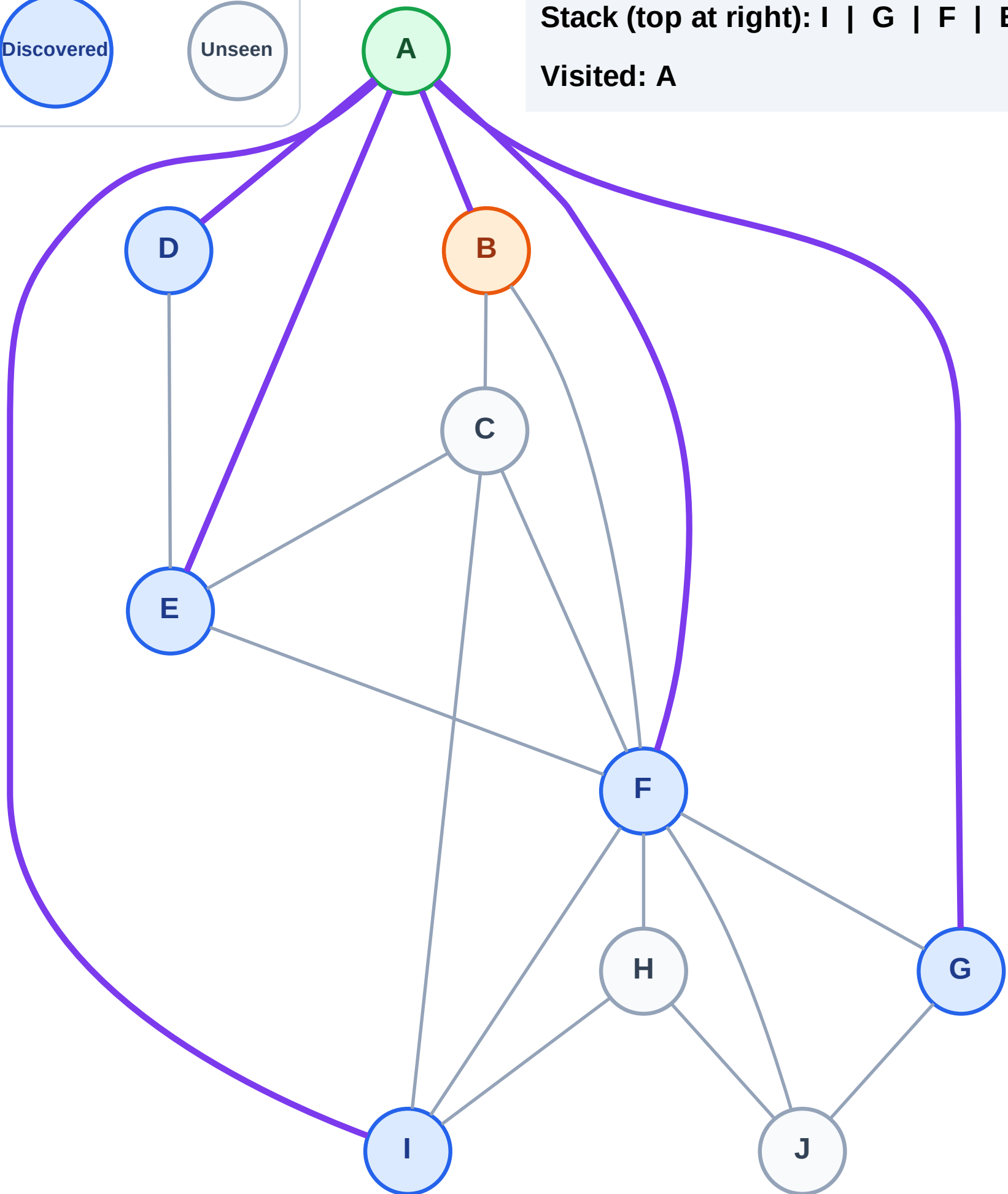
Processing

Discovered

Unseen

Stack (top at right): I | G | F | E | D

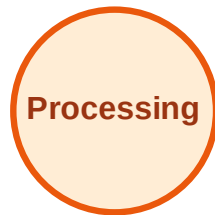
Visited: A



DFS Traversal

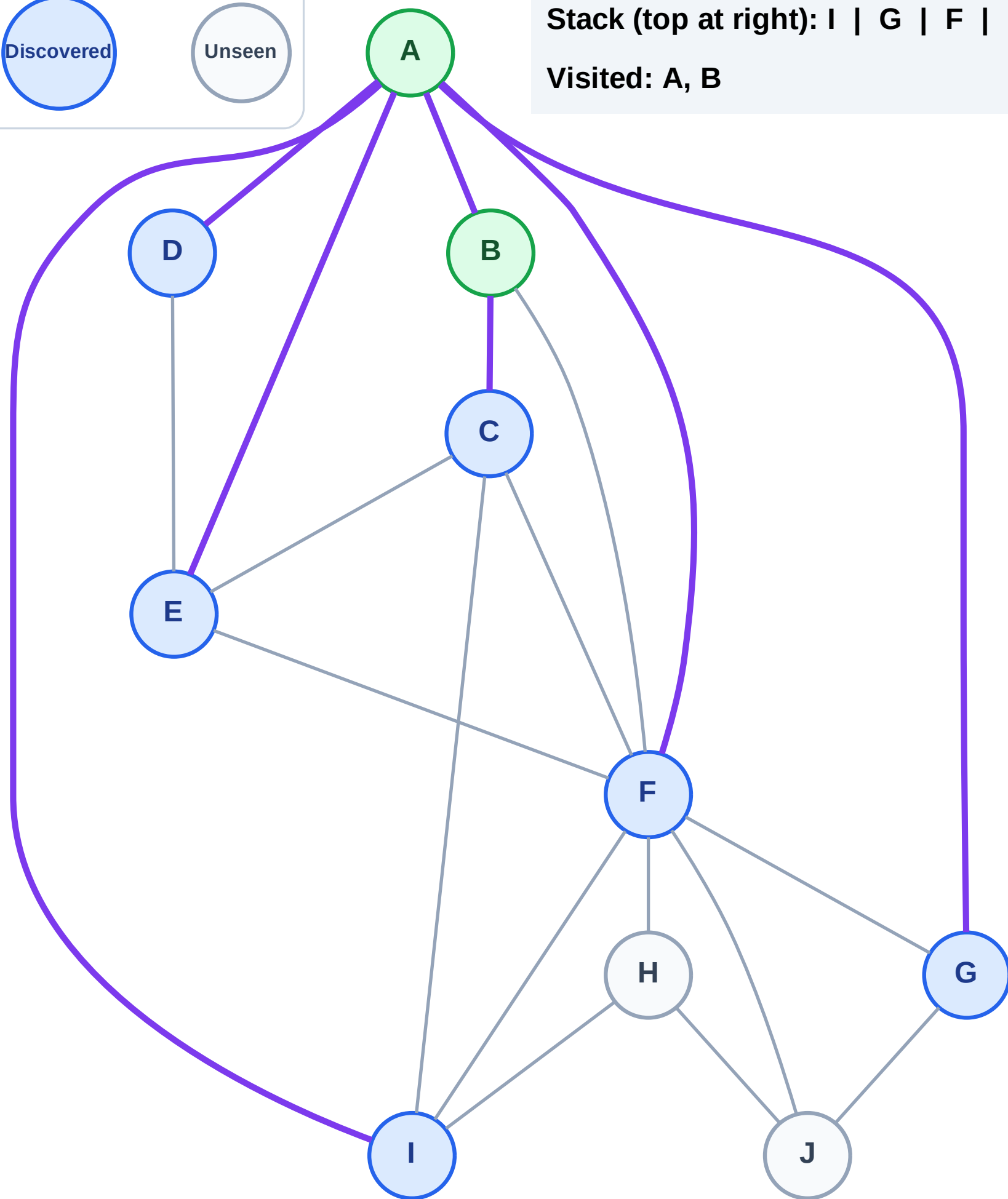
Step 5: Discover C from B

Legend



Stack (top at right): I | G | F | E | D | C

Visited: A, B



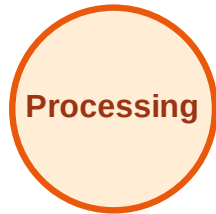
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



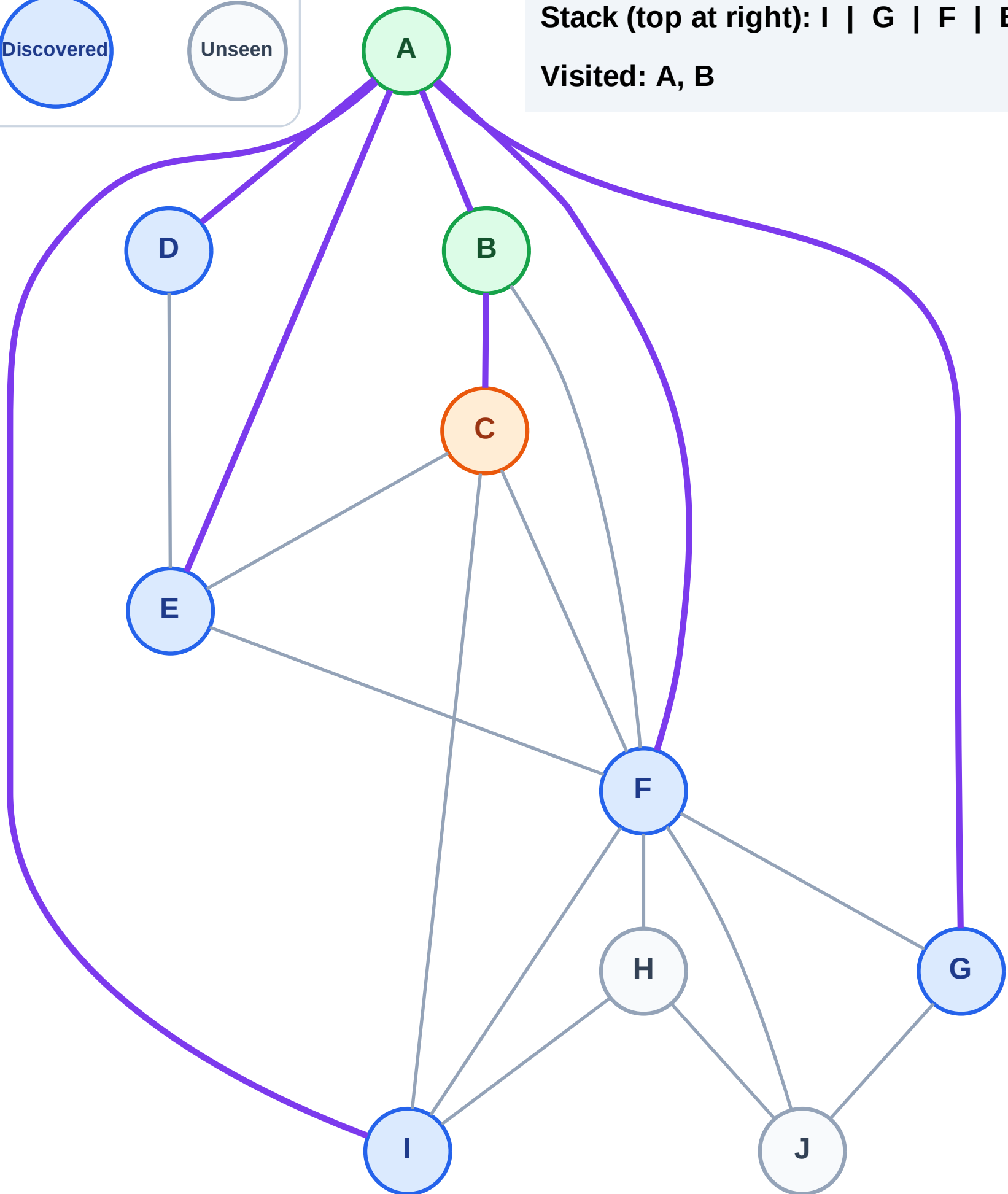
Discovered



Unseen

Stack (top at right): I | G | F | E | D

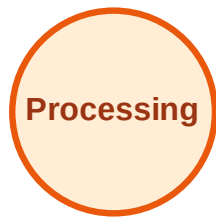
Visited: A, B



DFS Traversal

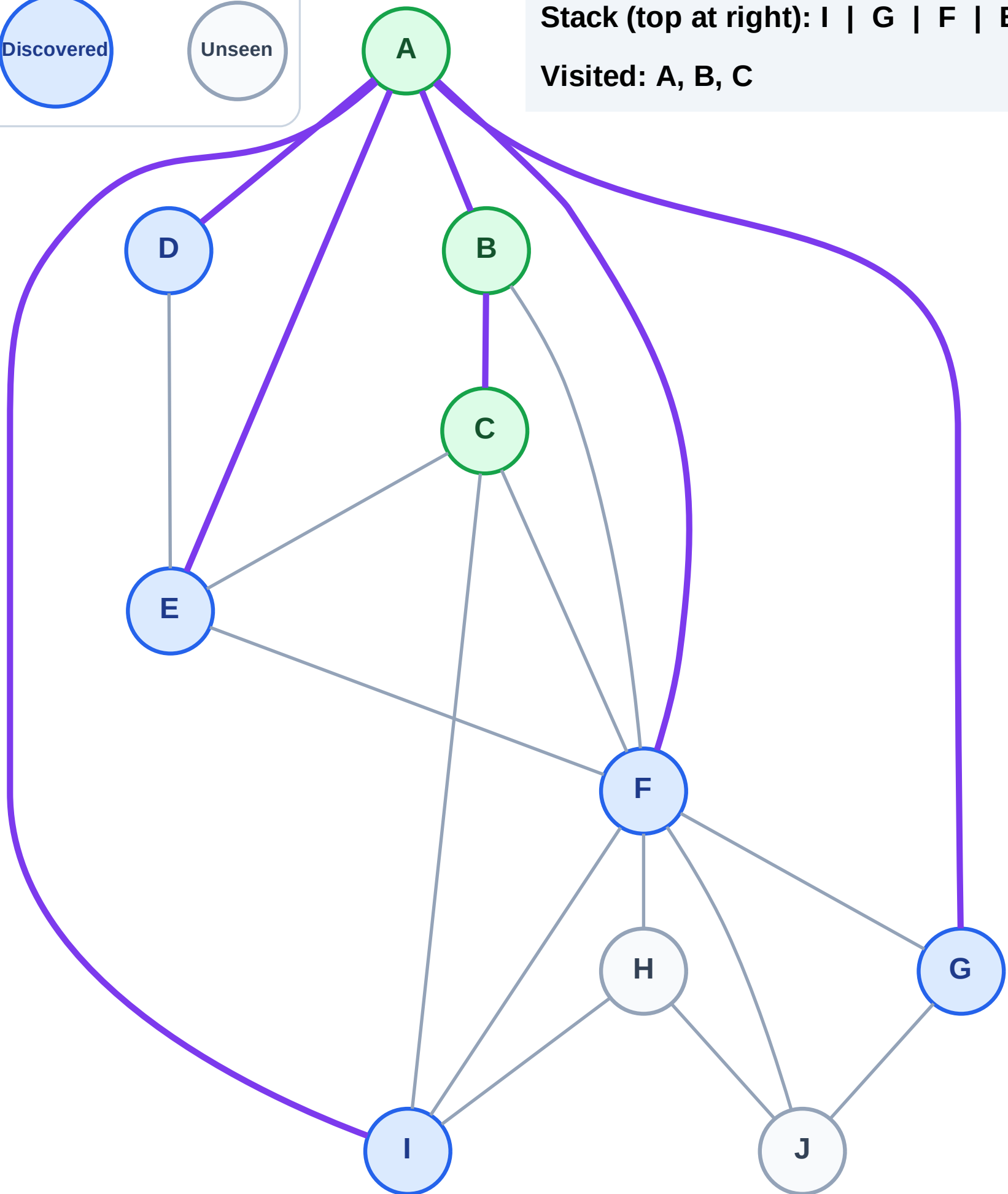
Step 7: No new neighbors from C

Legend



Stack (top at right): I | G | F | E | D

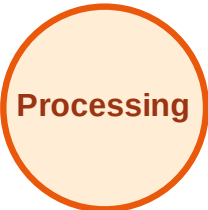
Visited: A, B, C



DFS Traversal

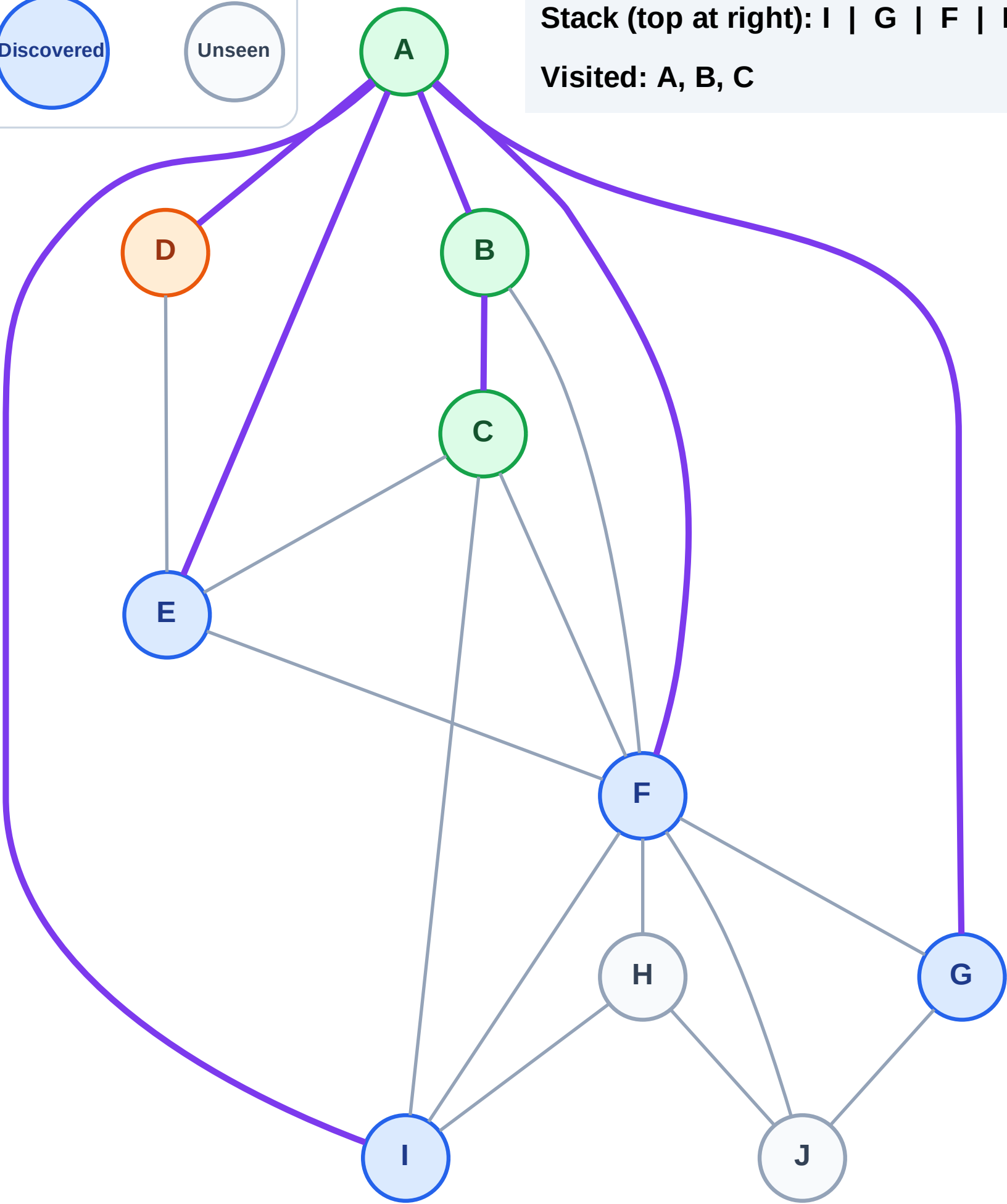
Step 8: Process node D

Legend



Stack (top at right): I | G | F | E

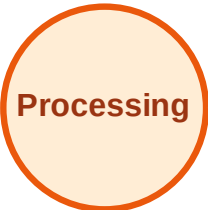
Visited: A, B, C



DFS Traversal

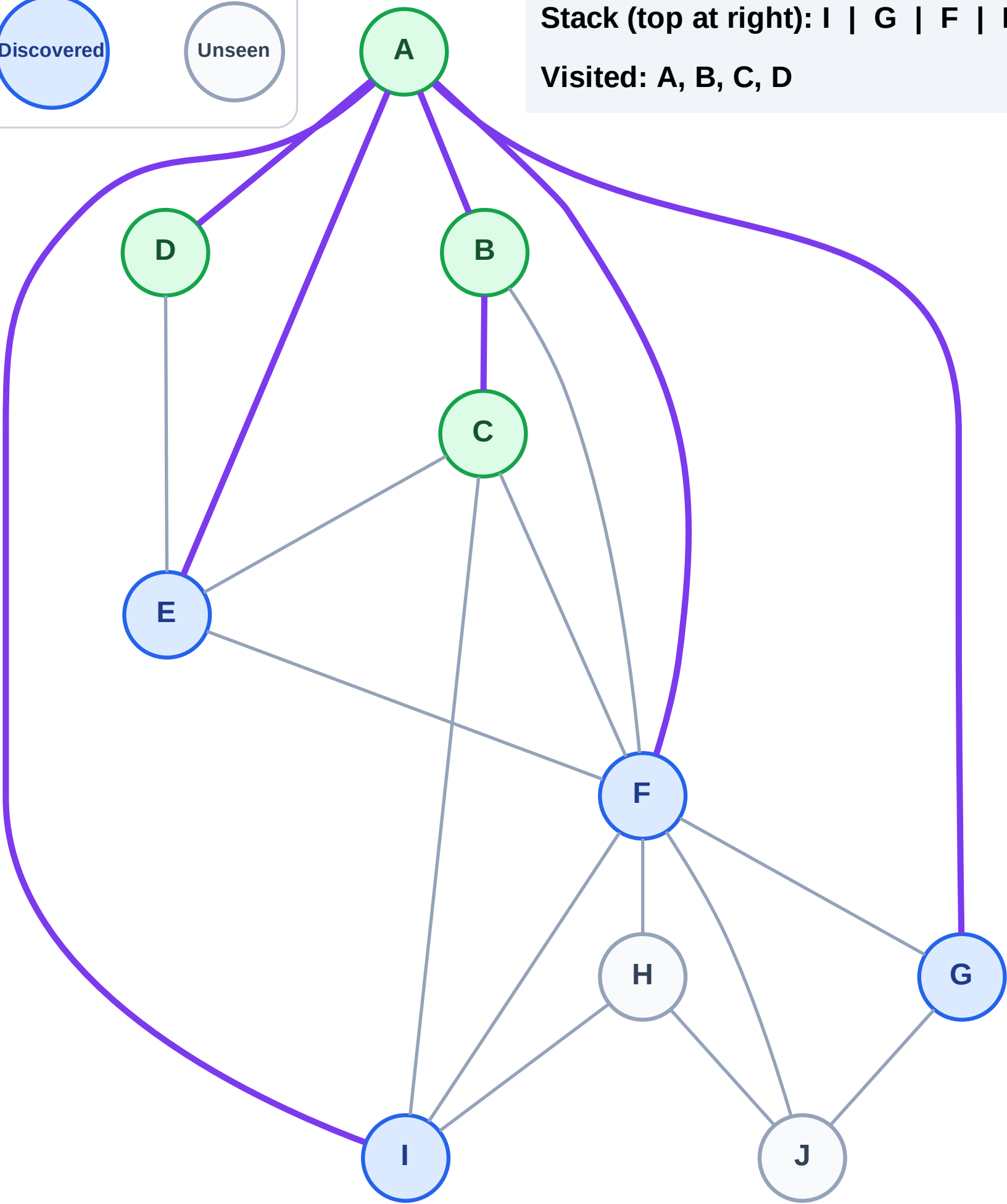
Step 9: No new neighbors from D

Legend



Stack (top at right): I | G | F | E

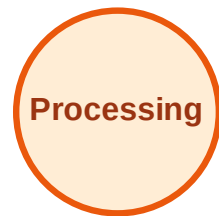
Visited: A, B, C, D



DFS Traversal

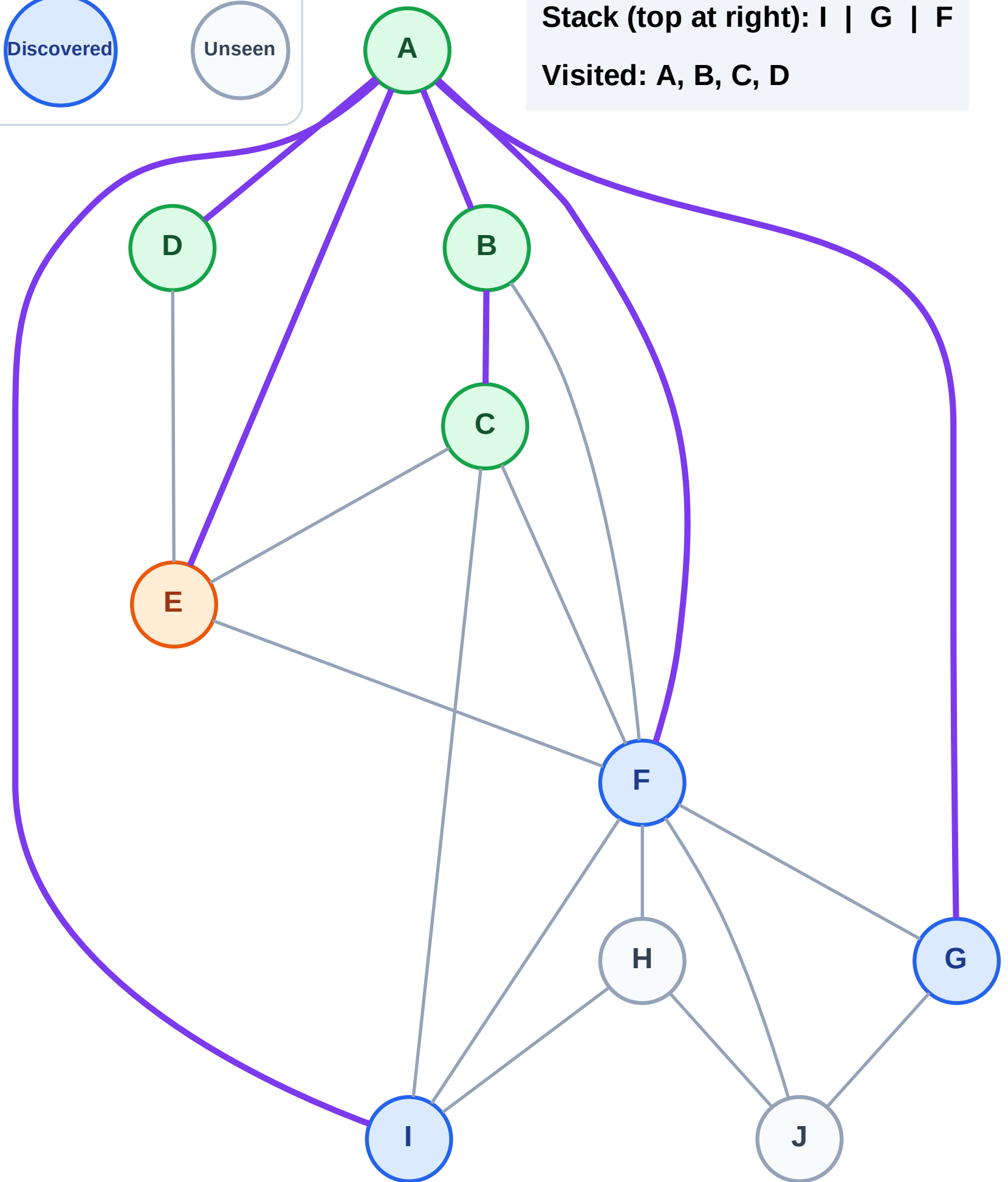
Step 10: Process node E

Legend



Stack (top at right): I | G | F

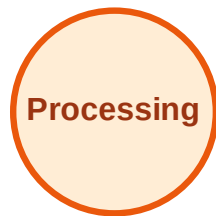
Visited: A, B, C, D



DFS Traversal

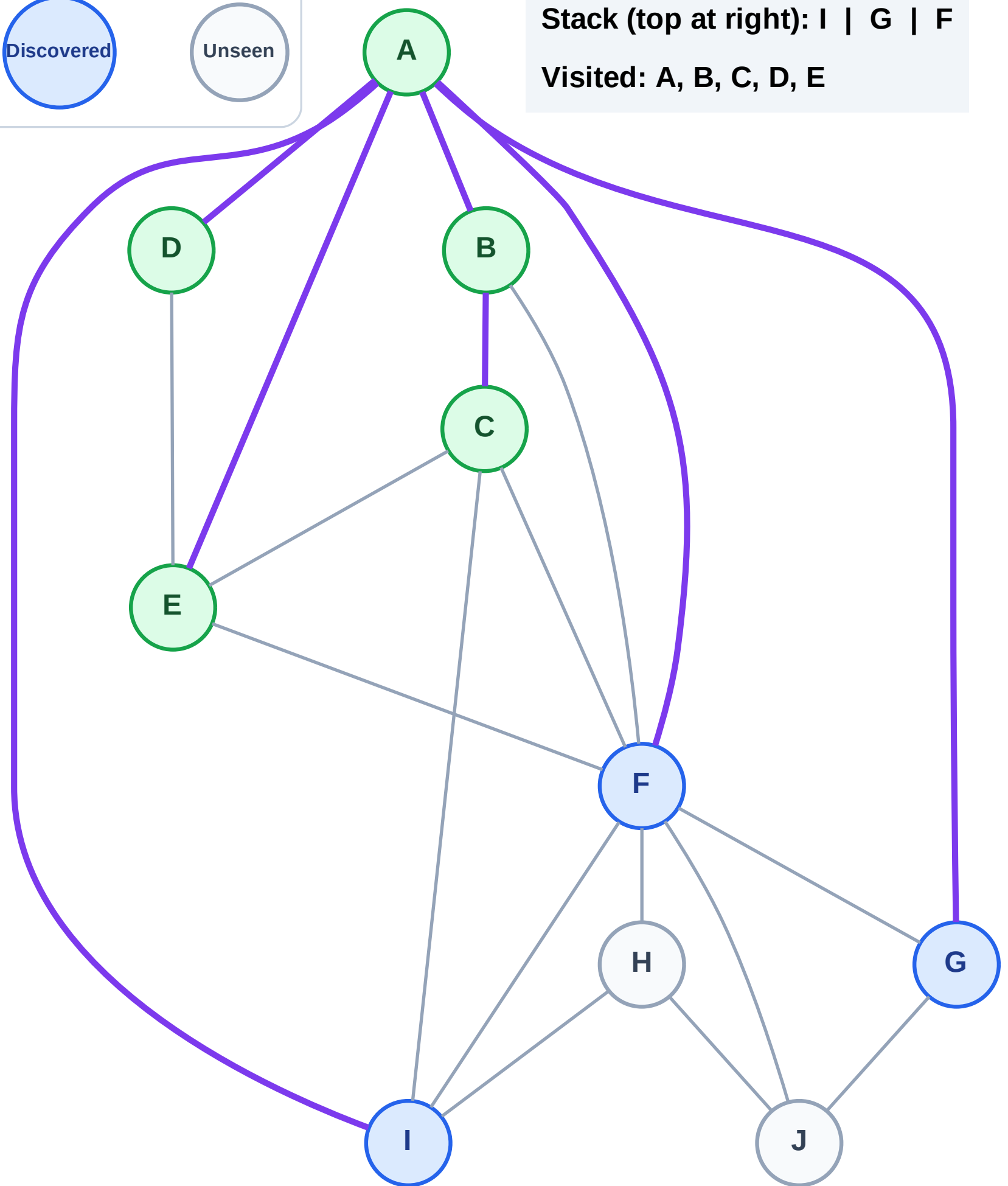
Step 11: No new neighbors from E

Legend



Stack (top at right): I | G | F

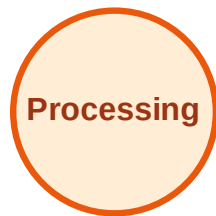
Visited: A, B, C, D, E



DFS Traversal

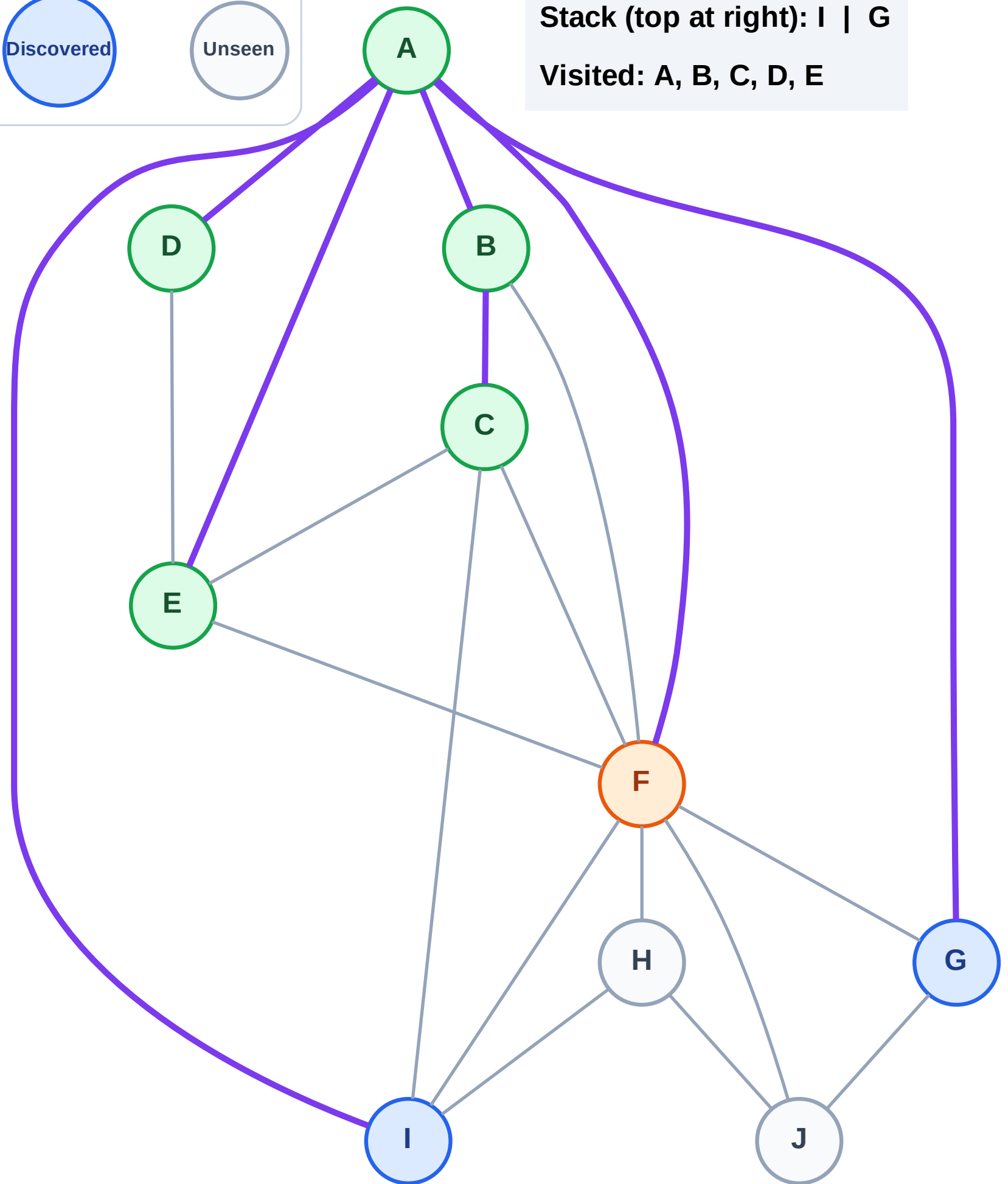
Step 12: Process node F

Legend



Stack (top at right): I | G

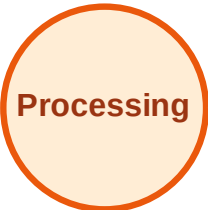
Visited: A, B, C, D, E



DFS Traversal

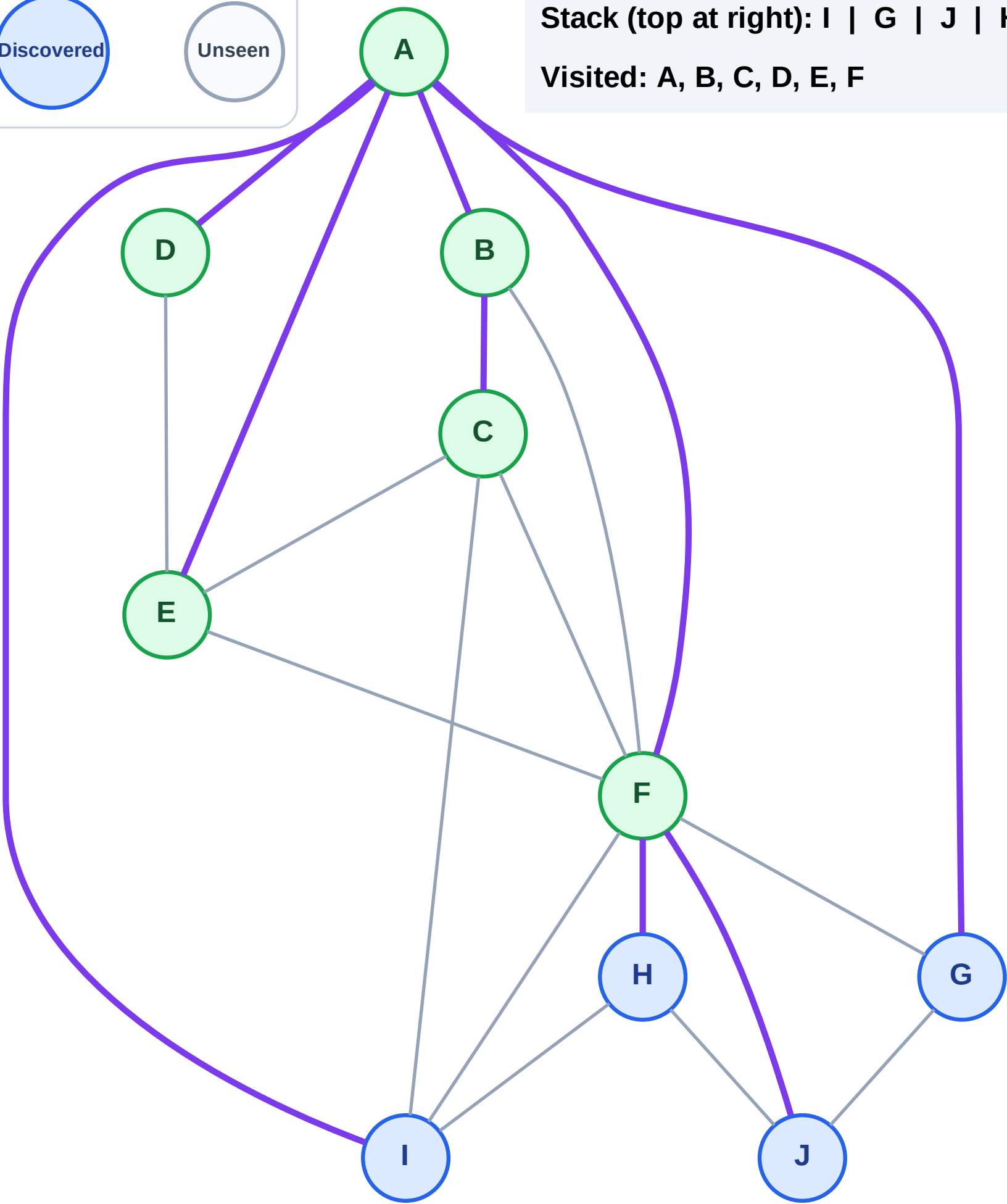
Step 13: Discover J, H from F

Legend



Stack (top at right): I | G | J | H

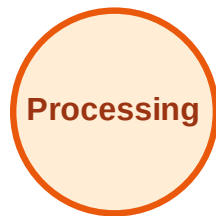
Visited: A, B, C, D, E, F



DFS Traversal

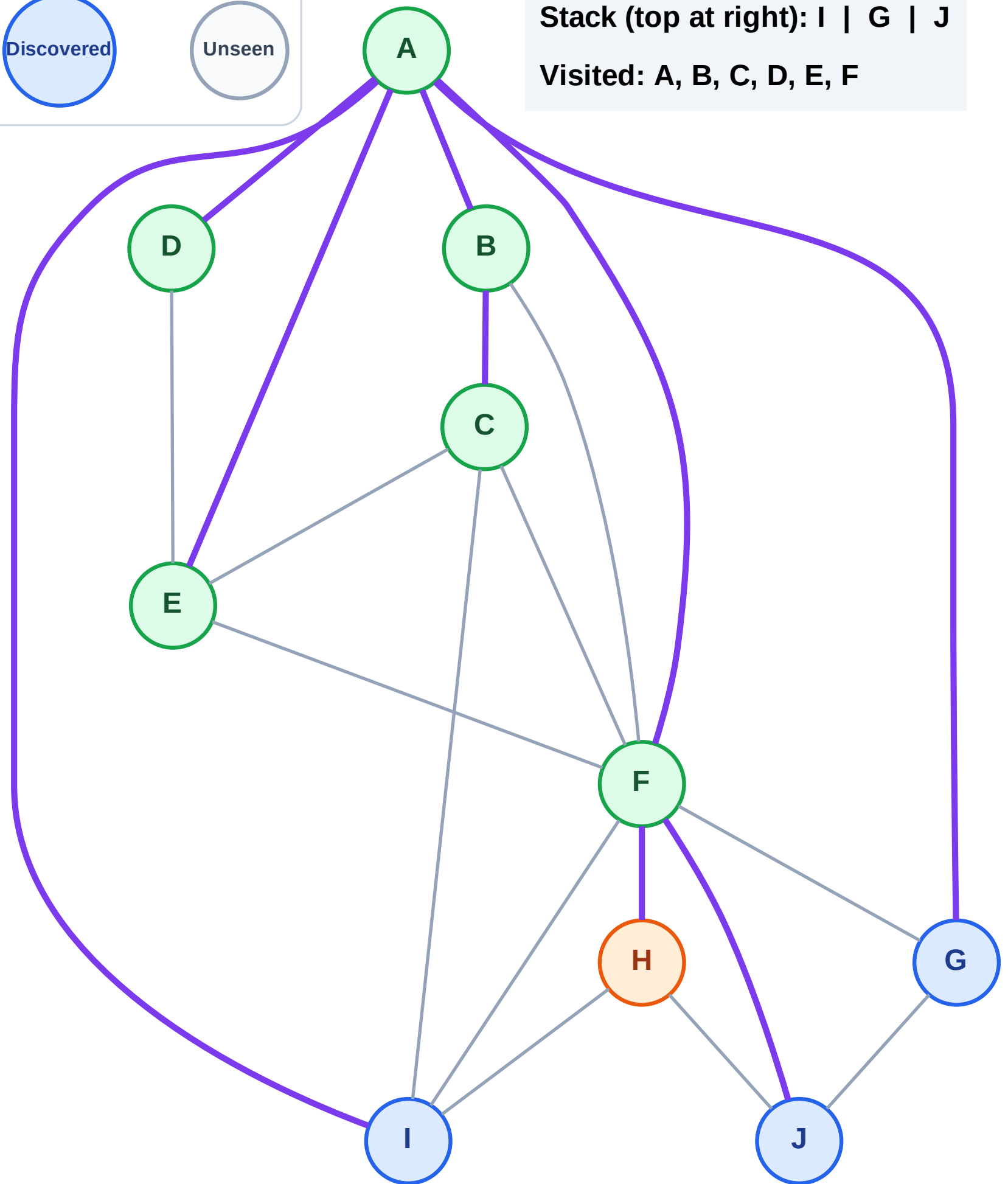
Step 14: Process node H

Legend



Stack (top at right): I | G | J

Visited: A, B, C, D, E, F



DFS Traversal

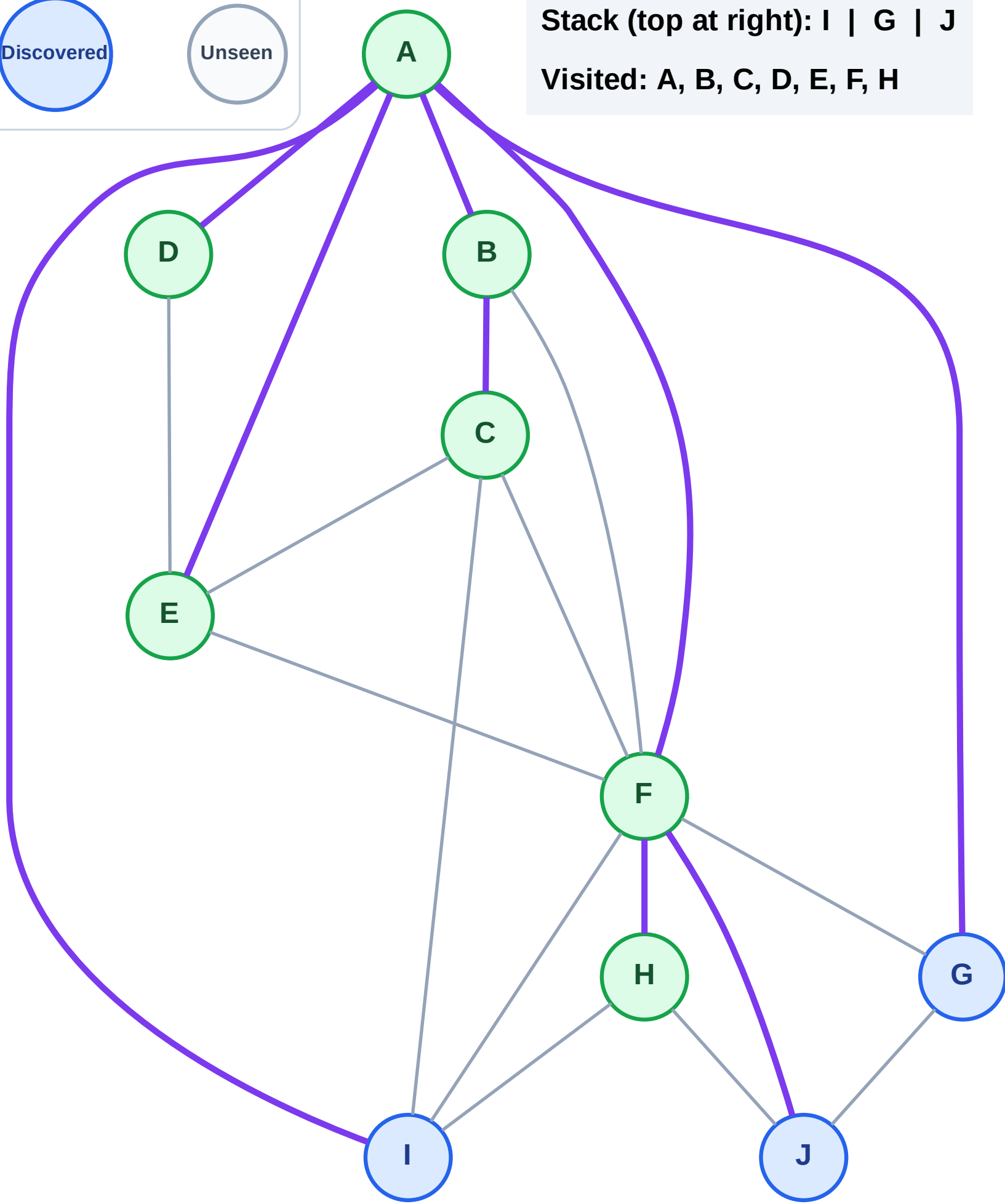
Step 15: No new neighbors from H

Legend



Stack (top at right): I | G | J

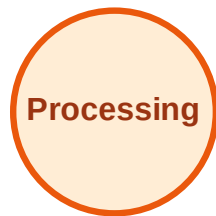
Visited: A, B, C, D, E, F, H



DFS Traversal

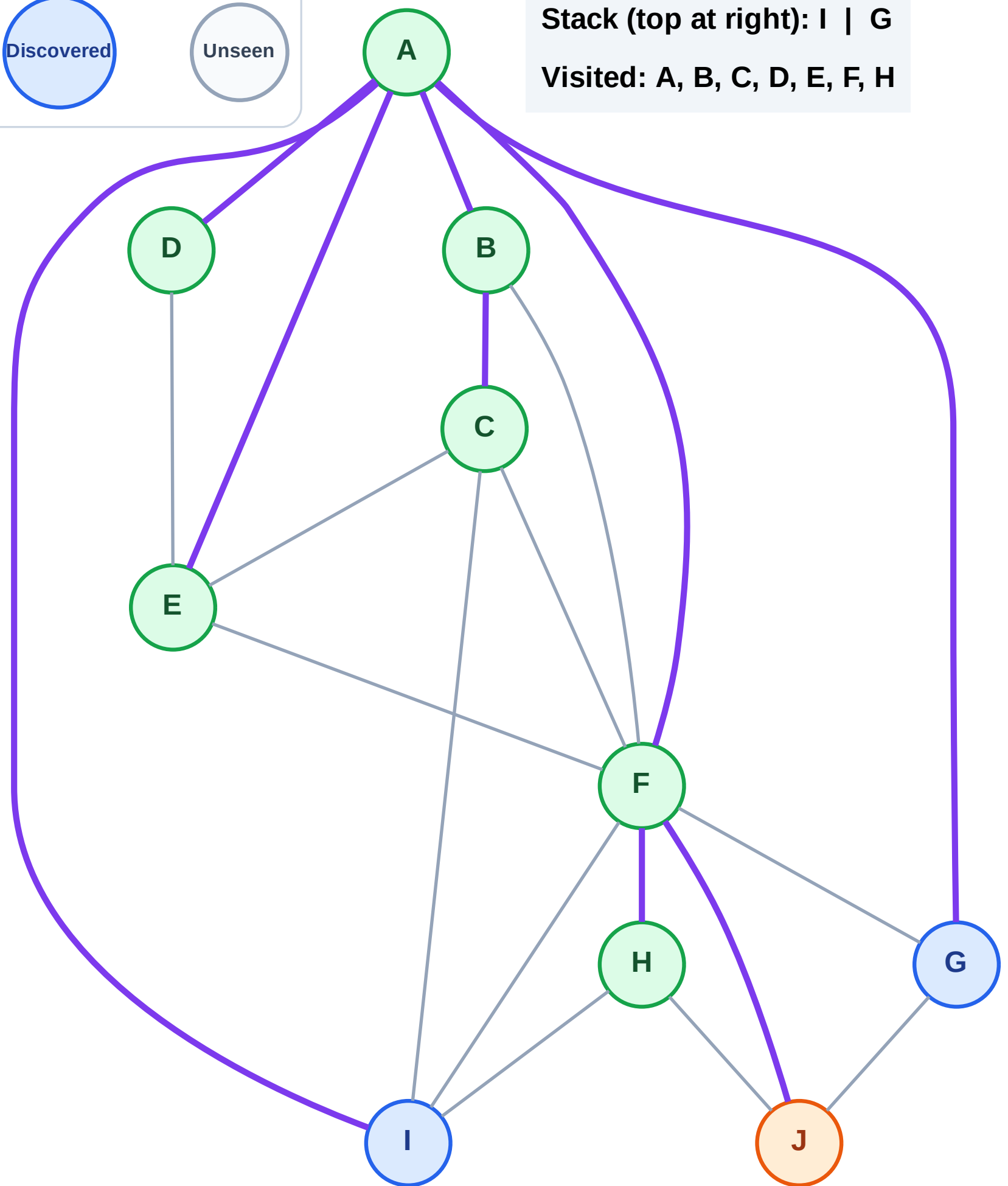
Step 16: Process node J

Legend



Stack (top at right): I | G

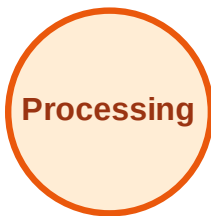
Visited: A, B, C, D, E, F, H



DFS Traversal

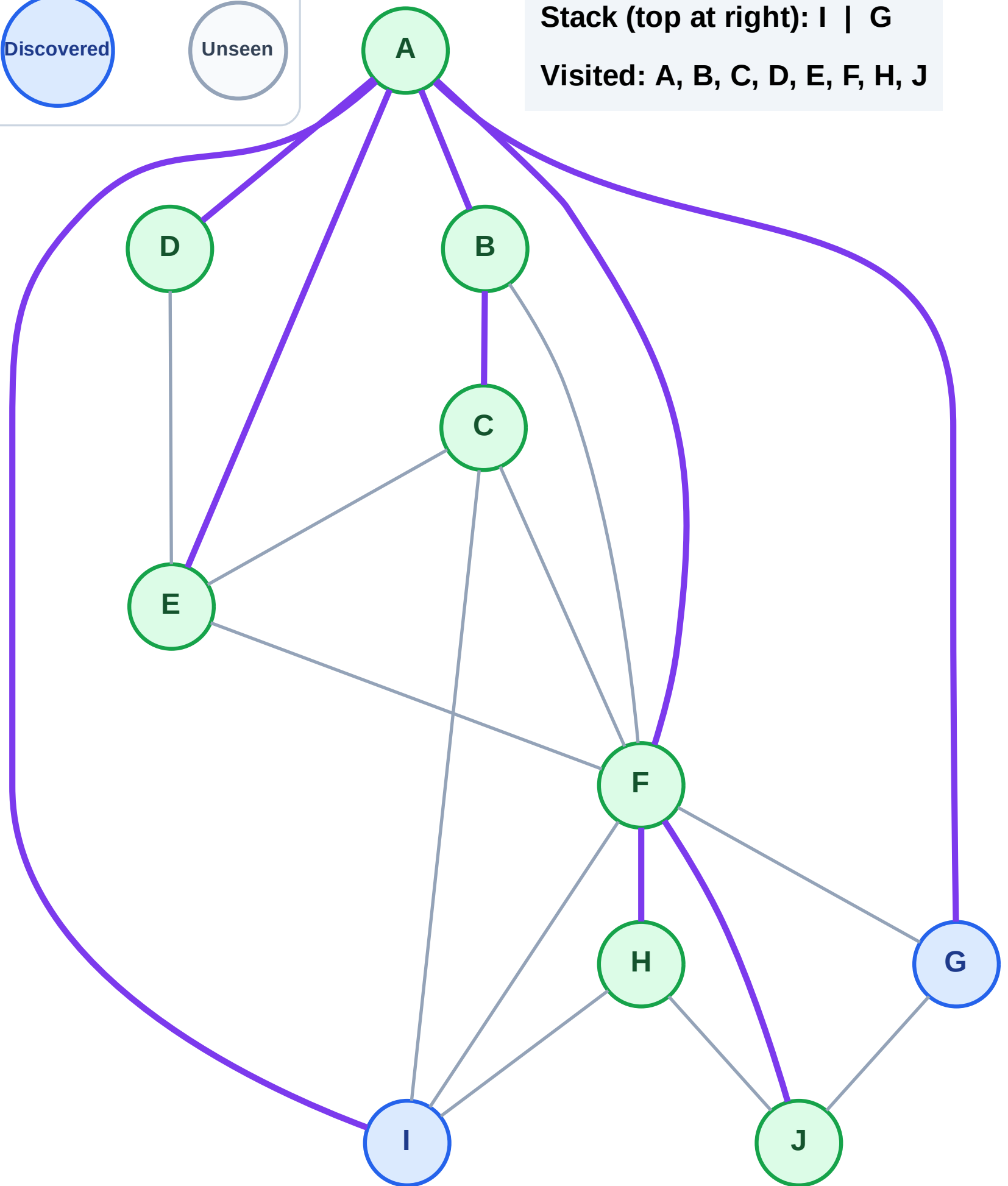
Step 17: No new neighbors from J

Legend



Stack (top at right): I | G

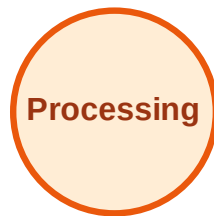
Visited: A, B, C, D, E, F, H, J



DFS Traversal

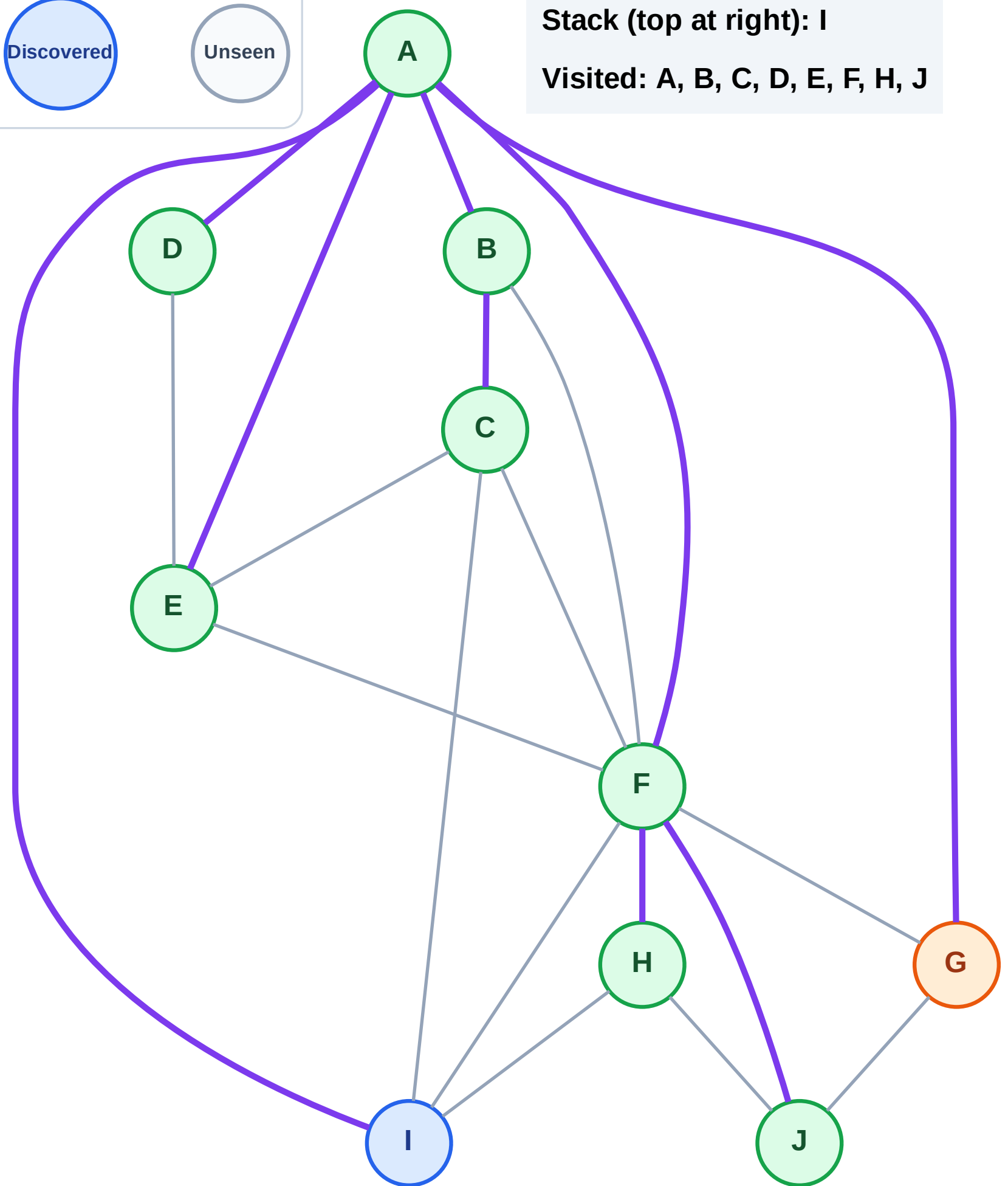
Step 18: Process node G

Legend



Stack (top at right): I

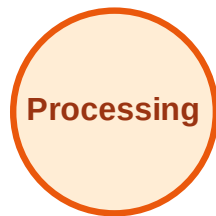
Visited: A, B, C, D, E, F, H, J



DFS Traversal

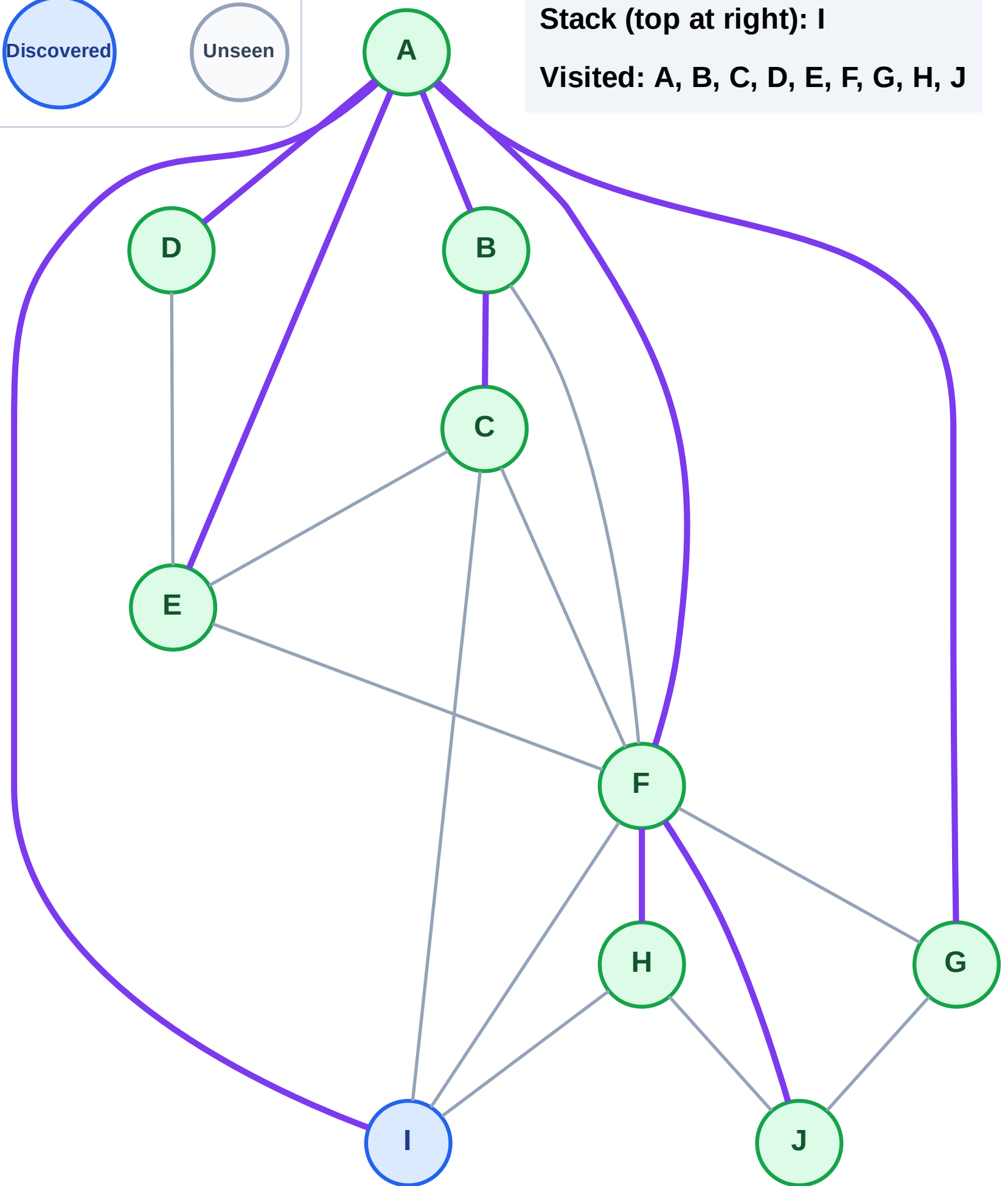
Step 19: No new neighbors from G

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

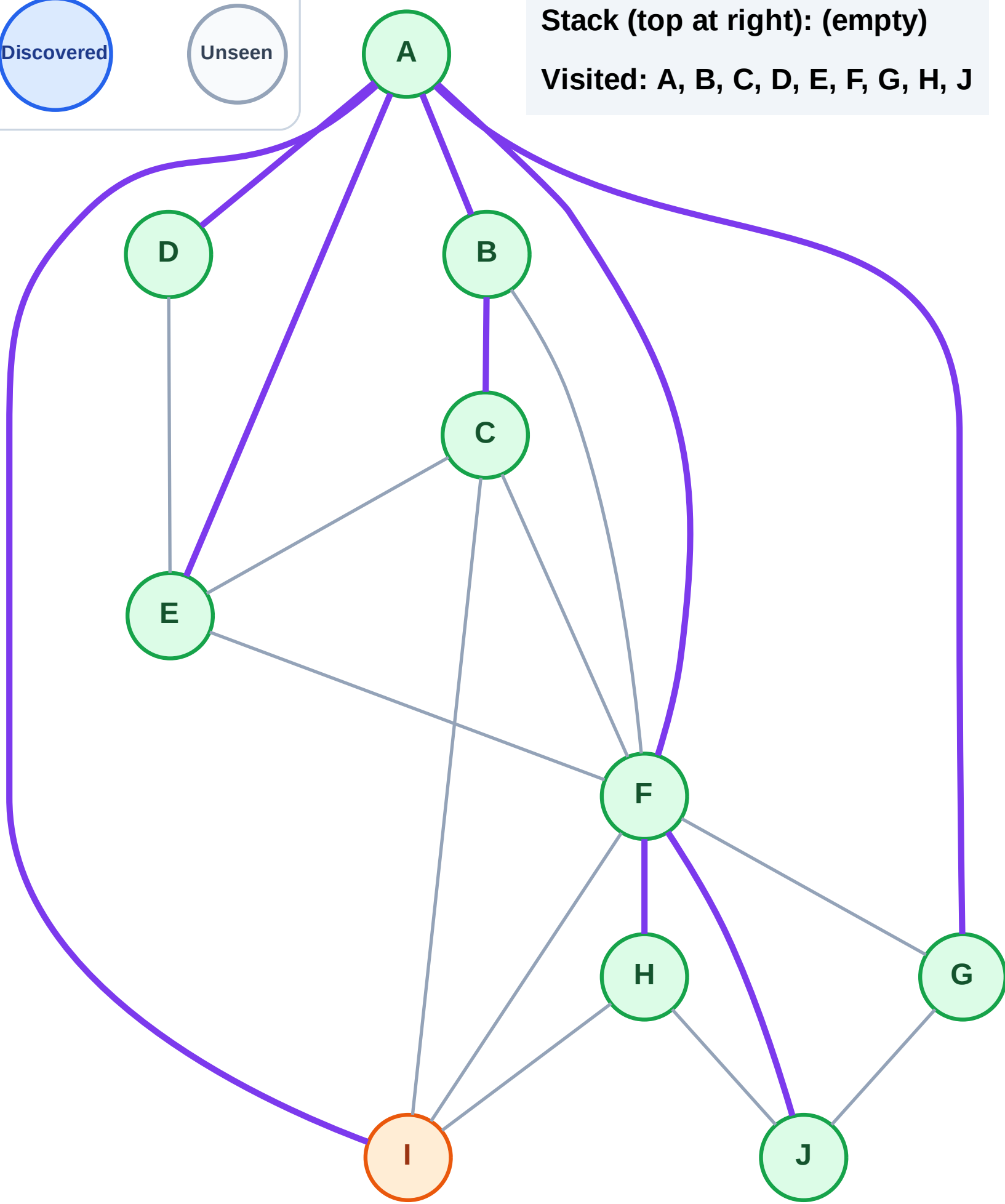
Step 20: Process node I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



Step 21: No new neighbors from I

Step 21: No new neighbors from I

